INSTRUCTIONS

GEK-36448

AC ELECTRIC ACTUATOR

GENERAL ELECTRIC

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These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company.

AC ELECTRIC ACTUATOR

INTRODUCTION

The General Electric AC Electric Actuator is a high performance servo device used to position process controlling elements such as dampers and vanes. It utilizes solid-state electronics, linear output motion, rugged construction and a number of unique features to provide accurate and reliable positioning for a wide range of power generating and process applications.

Each modulating actuator consists of one Mechanical Output Unit (MU) and one Electronic Control Unit (ECU). Each non-modulating actuator consists of one Mechanical Output Unit (MU) and a Reversing Controller Panel Unit (CPU).

NOTE

The first part of this instruction book will cover the modulating type actuators (servo system). The second part will cover the non-modulating type (on-off).

Other components such as a 3-phase supply transformer are sometimes furnished. Both the number and arrangement of auxiliary components vary with the requirements of the particular application. Therefore, it is necessary to refer to the Summary and diagrams provided for each individual application to determine what material is included and what connections should be used. The diagrams shown in this book are for illustrative purposes only: they are not intended to apply to all installations.

RECEIVING, HANDLING AND STORAGE

RECEIVING AND HANDLING

Immediately upon receipt, the equipment should be carefully unpacked to avoid damaging the apparatus. Particular care should be exercised to prevent small parts being mislaid or thrown away in the packing material.

As soon as the equipment is unpacked, it should be examined for any damage that might have been sustained in transit. If injury or rough handling is evident, a damage claim should be filed immediately with the transportation company and the nearest General Electric Sales Office should be notified promptly.

STORAGE

If the equipment is not to be used as soon as it is unpacked, it should be stored in a clean, dry place and protected from accidental damage. Particular care should be exercised to avoid storing the equipment in locations where construction work is in progress. Maximum storage temperature range must not exceed -50°C to +120°C.

DESCRIPTION

COMPONENTS

MECHANICAL UNIT (Figures 1A, 1B, 2)

The Mechanical Unit may be referred to as the "muscle" part of the system. It features a high-slip 3-phase AC induction motor with overload protection, eliminating brushes and commutators.

A single-phase AC disc brake is used for accurate positioning in Manual Mode and for protection of the driven element and/or the MU when in Auto Mode if the MU is driven past either of the extreme end-of-travel limit switches.

There are four basic MUs. See specifications, Figure 2. Low efficiency gearing will hold position against back-driving forces, allowing the motor to be energized only when the actuator is required to move to a new position.

Linear motion minimizes mechanical distortion.

The position feedback mechanism uses an LVDT (linear variable differential transformer) and a long characterizing cam for an accurate feedback signal.

An overforce limiter mechanically and electrically limits over-travel due to excessive forces in either direction of travel.

A tachometer, belt driven from the input shaft of the jack, provides stable operation at various motor speeds and actuator loads.

The demodulator assembly converts the AC signal from the output of the LVDT to a DC level, which is fed back to the ECU as the actuator position signal.

There are three sets of limit switches: RNLS and ENLS - near-end-of travel (actuates circuit to slow

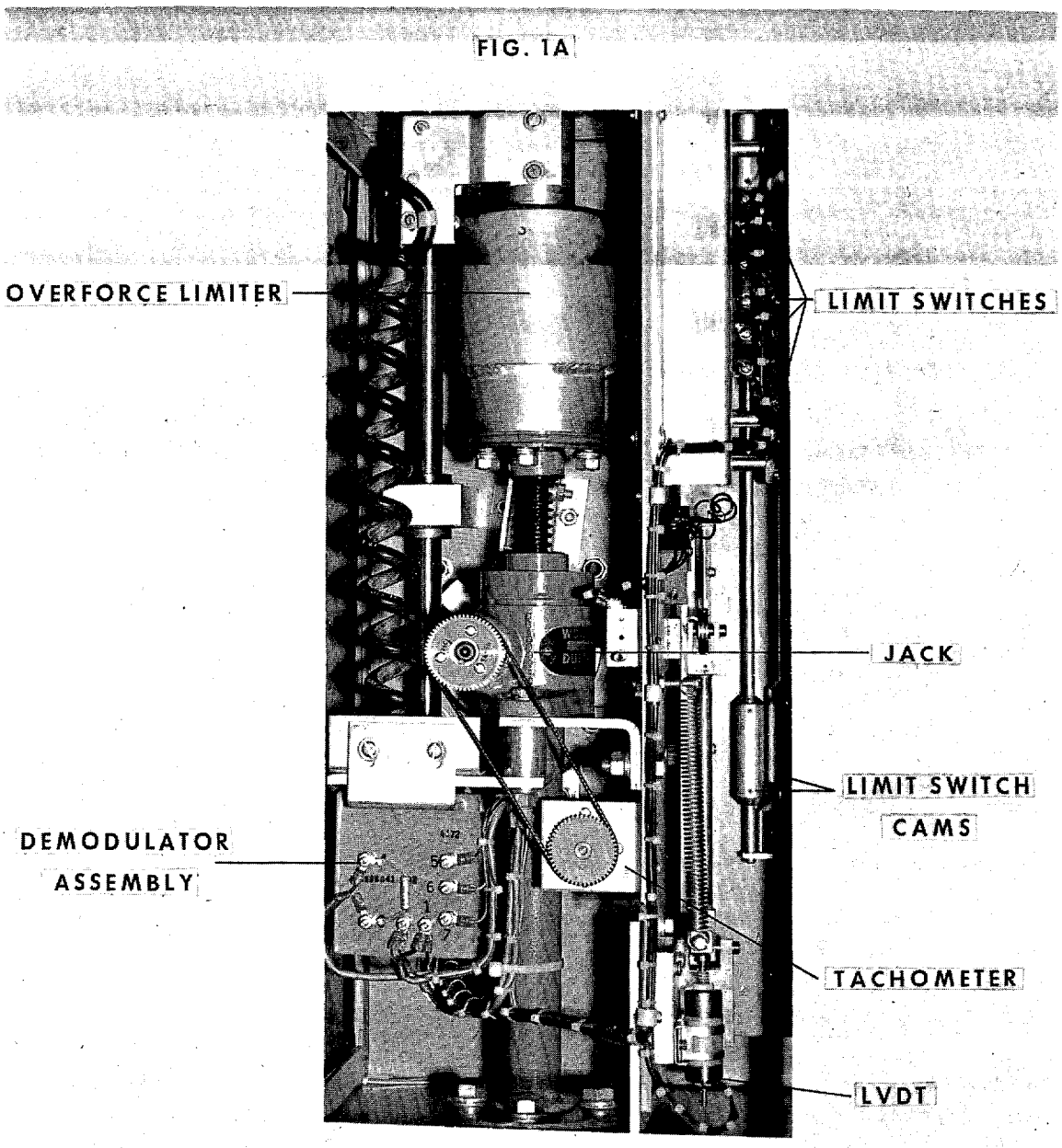


Figure 1A. Front View of MU

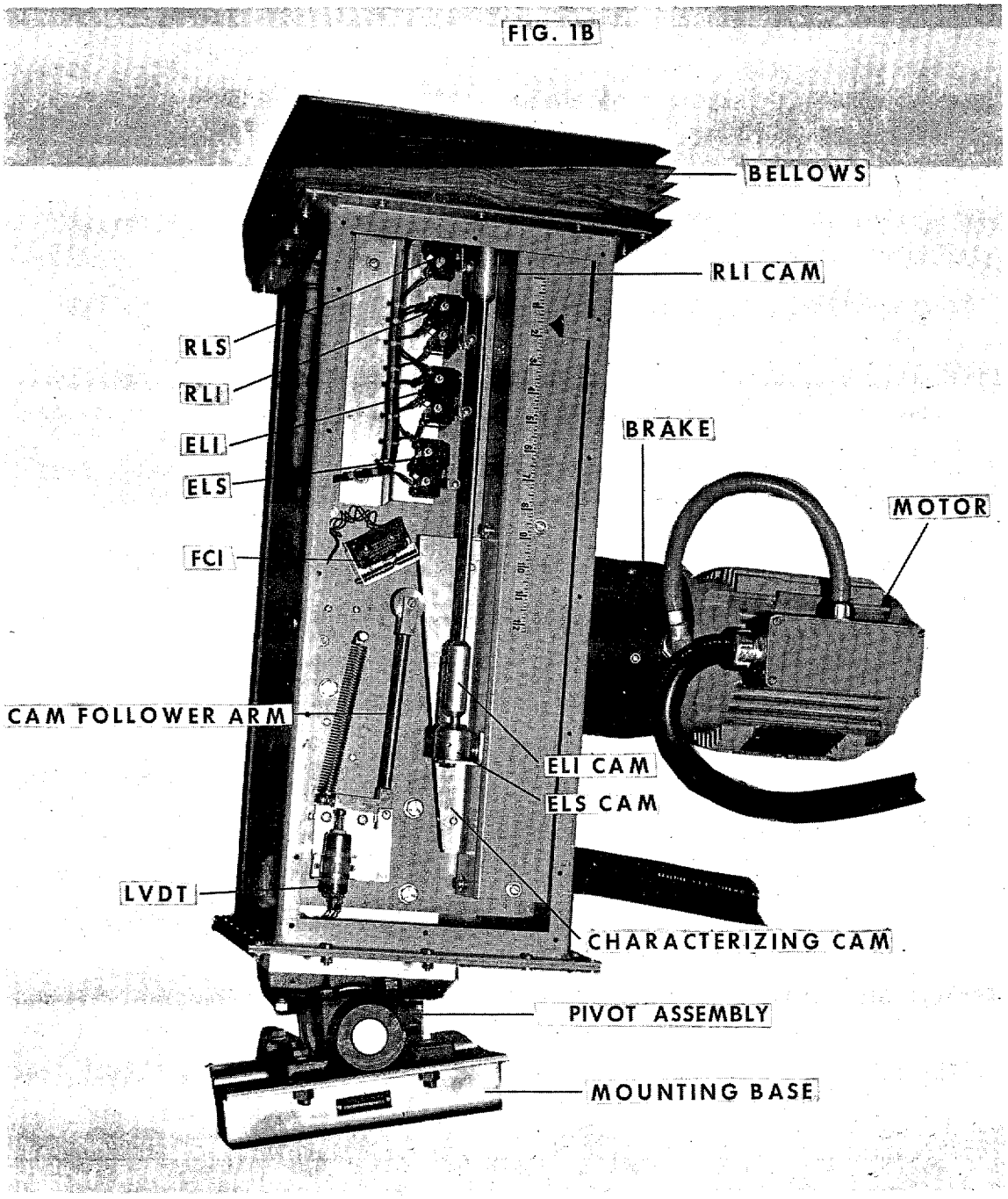
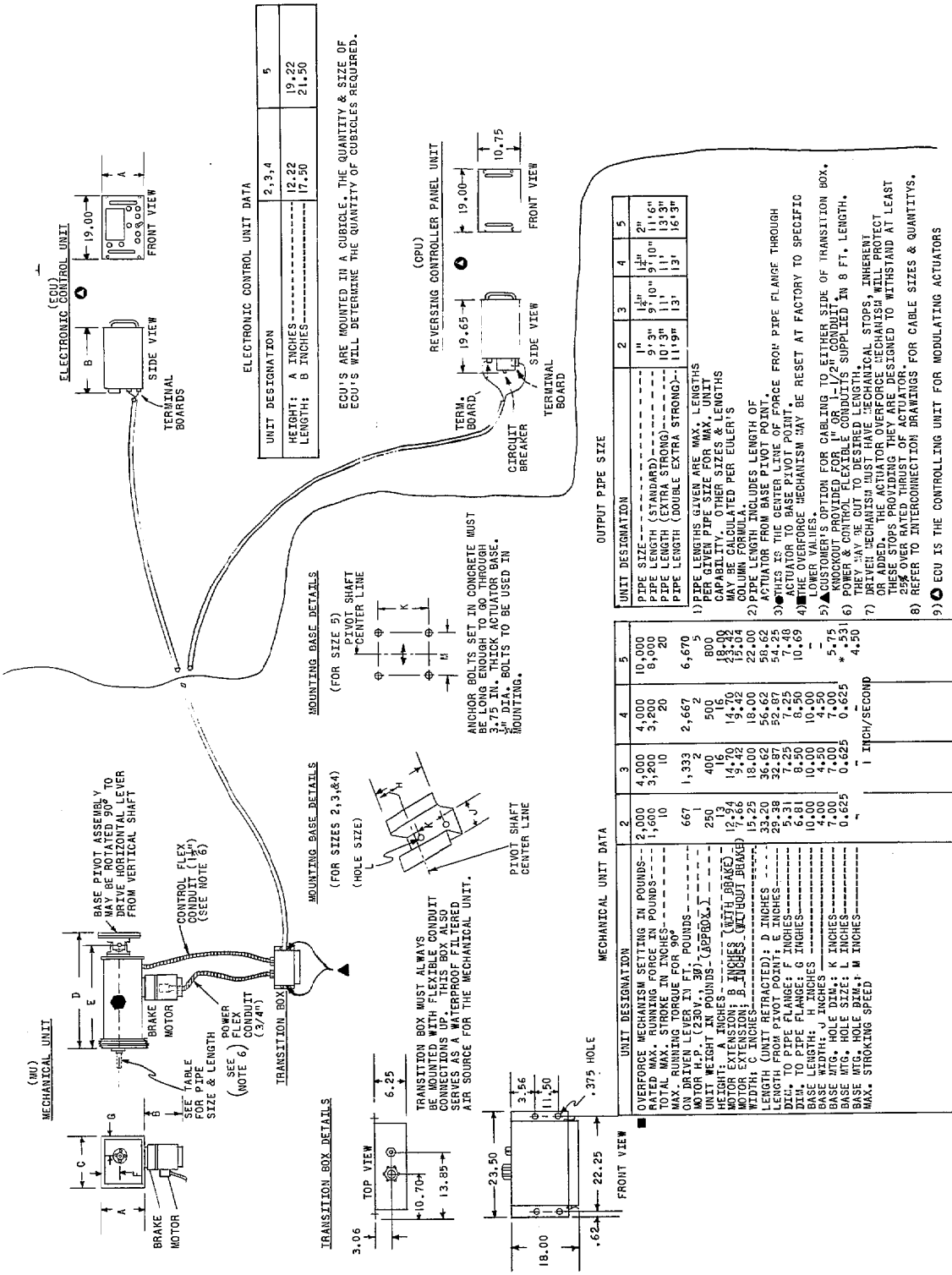


Figure 1B. Side View of MU



ELECTRONIC CONTROL UNIT DATA			
UNIT DESIGNATION	2,3,4	5	
HEIGHT: A INCHES	12.22	19.22	
LENGTH: B INCHES	17.50	21.50	

MECHANICAL UNIT DATA					
UNIT DESIGNATION	2	3	4	5	
OVERFORCE MECHANISM SETTING IN POUNDS-	2,000	4,000	4,000	10,000	
RATED MAX. FORCE IN POUNDS-	1,600	3,200	3,200	8,000	
TOTAL MAX. STROKE IN INCHES-	10	10	20	20	
MAX. RUNNING TORQUE FOR 90° ON DRIVEN LEVER IN FT. POUNDS-	667	1,333	2,667	6,670	5
MOTOR H.P. (230V., 3Ø)-	1	2	2		
UNIT WEIGHT IN POUNDS-(APPROX.)	250	400	500	800	
HEIGHT: A INCHES-	13	16	16	18.00	
MOTOR EXTENSION: B INCHES (WITH BRAKE)-	12.94	14.70	14.70	23.42	
MOTOR EXTENSION: B INCHES (WITHOUT BRAKE)	7.66	9.42	9.42	15.04	
WIDTH: C INCHES-	15.25	18.00	18.00	22.00	
LENGTH (UNIT RETRACTED): D INCHES-	33.20	36.62	56.62	56.62	
LENGTH FROM PIVOT POINT: E INCHES-	29.38	32.87	52.87	54.25	
DIM. TO PIPE FLANGE: F INCHES-	5.31	7.25	7.25	7.48	
DIM. TO PIPE FLANGE: G INCHES-	6.81	8.50	8.50	10.69	
BASE LENGTH: H INCHES-	10.00	10.00	10.00	-	
BASE WIDTH: J INCHES-	4.00	4.50	4.50	-	
BASE MTG. HOLE DIM.: K INCHES-	7.00	7.00	7.00	5.75	
BASE MTG. HOLE SIZE: L INCHES-	0.625	0.625	0.625	*.531	
BASE MTG. HOLE DIM.: M INCHES-	-	-	-	4.50	
MAX. STROKING SPEED			1 INCH/SECOND		

OUTPUT PIPE SIZE				
UNIT DESIGNATION	2	3	4	5
PIPE SIZE-	1"	1¼"	½"	2"
PIPE LENGTH (STANDARD)-	9'3"	9'10"	8'10"	11'6"
PIPE LENGTH (EXTRA STRONG)-	10'3"	11'	11'	13'3"
PIPE LENGTH (DOUBLE EXTRA STRONG)-	11'9"	13'	13'	16'3"

1) PIPE LENGTHS GIVEN ARE MAX. LENGTHS PER GIVEN PIPE SIZE FOR MAX. UNIT CAPABILITY. OTHER SIZES & LENGTHS MAY BE CALCULATED PER EULER'S COLUMN FORMULA.
2) PIPE LENGTH INCLUDES LENGTH OF ACTUATOR FROM BASE PIVOT POINT.
3) ● THIS IS THE CENTER LINE OF FORCE FROM PIPE FLANGE THROUGH ACTUATOR TO BASE PIVOT POINT.
4) ■ THE OVERFORCE MECHANISM MAY BE RESET AT FACTORY TO SPECIFIC LOWER VALUES.
5) ▲ CUSTOMER'S OPTION FOR CABLING TO EITHER SIDE OF TRANSITION BOX. KNOCKOUT PROVIDED FOR 1" OR 1-1/2" CONDUIT.
6) POWER & CONTROL FLEXIBLE CONDUITS SUPPLIED IN 8 FT. LENGTH. THEY MAY BE CUT TO DESIRED LENGTH.
7) DRIVEN MECHANISM MUST HAVE MECHANICAL STOPS, INHERENT OR ADDED. THE ACTUATOR OVERFORCE MECHANISM WILL PROTECT THESE STOPS PROVIDING THEY ARE DESIGNED TO WITHSTAND AT LEAST 25% OVER RATED THRUST OF ACTUATOR.
8) REFER TO INTERCONNECTION DRAWINGS FOR CABLE SIZES & QUANTITYS.
9) ○ ECU IS THE CONTROLLING UNIT FOR MODULATING ACTUATORS
CPU IS THE CONTROLLING UNIT FOR NON-MODULATING ACTUATORS

* CENTERS MAY BE SHIFTED ± 1/8".

Figure 2. Outline/Specification

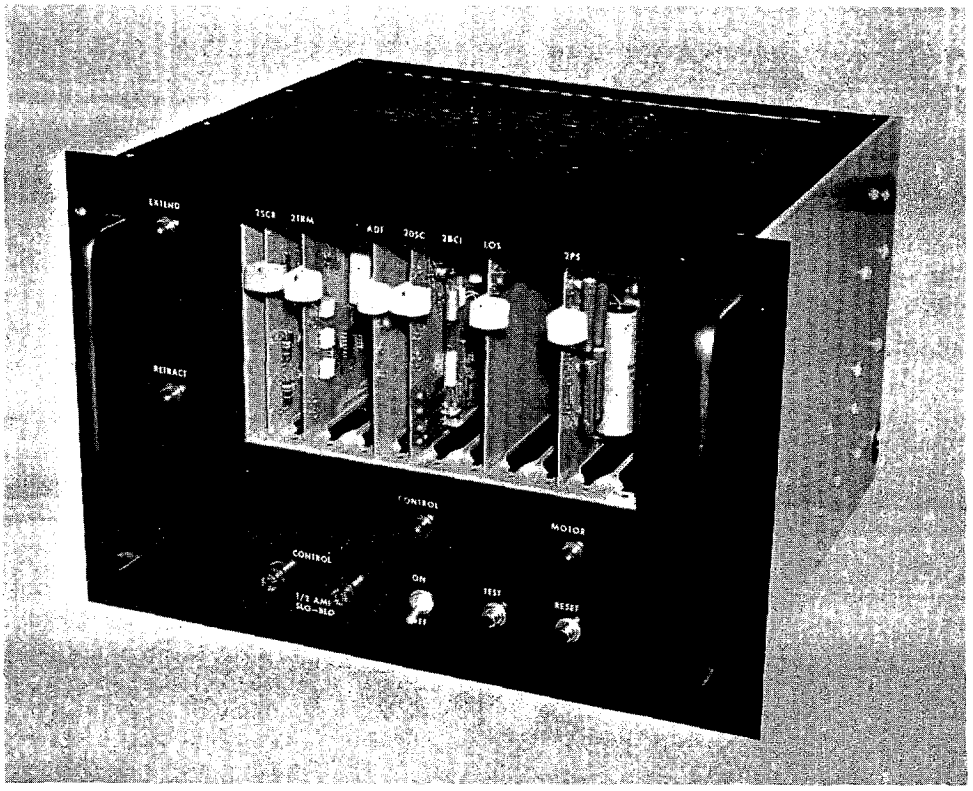


Figure 3. Front View ECU

speed of motor), RLI and ELI - end-of-travel (provides customer indication and deenergizes motor control circuit), and RLS and ELS - extreme-end-of-travel (applies brake and transfers mode of operation to Manual).

ELECTRONIC CONTROL UNIT (Figure 3)

The Electronic Control Unit contains all the control and power circuitry. It receives the input position demand signal and utilizes its circuitry to control the direction and speed of the MU. Almost all of the control circuitry is incorporated on highly reliable printed circuit cards. The cards contribute to easy, quick testing and maintenance.

There are two physically different sizes of ECUs, both of which are 19" rack mounted. There are three electrical sizes of ECUs (see specifications, Figure 2). The back panel contains all the termination points for control and power wiring and is hinged for easy access for servicing.

The ECUs are usually mounted in a cubicle remote from the MU.

CONTROLLER PANEL UNIT

The Controller Panel Unit contains the control circuitry for the non-modulating type actuators. They are the standard 19" rack mounted units. Each CPU may contain one or two complete control circuits; therefore, one panel is capable of controlling two MUs.

CUBICLE

In addition to the ECUs, the cubicle contains associated ECU circuit breakers (one per ECU), a bank of motor series resistors (three per ECU) mounted on the rear door, and may also contain CPUs.

All cubicles are identical and each may be considered as a module because the quantity required per application will depend on the quantity and size of the ECUs and CPUs. The modules are then bolted together and shipped as a single unit. The majority of applications require no electrical connections between cubicles.

TRANSITION BOX

The transition box serves two purposes:

1. It contains terminal boards as a cable connection median between the ECU and MU. Two, 8 ft. lengths of flexible conduit (3/4" and 1 1/2") connect the transition box to the MU.

2. The 1 1/2" conduit not only contains the interconnecting wiring but also serves as a breathing port for the MU as the MU extends and retracts. A polyurethane filter is used to filter the air as it enters the transition box. This type of filter can be washed in soapy water and reused. The filter cleaning should not have to be done very often, even in the most severe atmosphere because the volume of air flowing through the filter is very low.

OPTIONS

1. 208/230 volt operation is normal. Any other voltage requires a step-up or step-down transformer.

2. ADF printed circuit card.

This circuit is used when demand signals are noisy. These noisy signals require the MU to constantly reposition, reducing the mechanical life of the MU.

3. LOS printed circuit card.

This circuit detects a loss of demand signal, transferring the actuator to Manual Mode.

SPECIFICATIONS

(Refer to Outline/Specification, Figure 2)

INSTALLATION

MOUNTING, LINKAGE AND LIMIT SWITCH CAM SETTINGS

Mount the MU as near as possible to the driven element per base mounting details in Figure 2. Always mount the MU so the base pivot plane corresponds with the movement of the elements driven arm. If a different base orientation is required, remove the four bolts holding the base assembly to the frame and jack flange. (Access to these four bolts is through the front cover. The bolts go through the jack flange into the 1" block of the base assembly.) Rotate the base assembly 90° and remount.

*Manually position the MU to one extreme limit position, preferably the extreme retract limit position, with the pointer-indicator at "0" inches as viewed through the side window. (Turning the manual crank counterclockwise will retract the MU, as viewed from motor end.) Be sure the RLI limit switch has just been depressed and a "click" heard from RLI when the pointer just reaches 0 inches. The cam is factory set and should not have to be readjusted.

Place the driven element to the corresponding extreme position and install the necessary linkage between the MU and the driven element. See Figure 2 for output pipe size. Adjust linkage so MU will correspond with the driven element.

Place the driven element in the opposite extreme position, extend by manually cranking the MU clockwise. If the full length of the travel of the MU is being utilized, the pointer indicator should indicate the MU is at its maximum position (10" or 20"). The ELI limit switch should have just been depressed and a "click" heard when the pointer just reached maximum travel. The ELI cam, like the RLI cam, is factory set and should not have to be readjusted.

RNLS and ENLS cams were set at the factory at some predetermined load. Their final adjustment is to be made when each actuator system is completely installed and in the Auto Mode of control. Their setting will be determined by the % of actuator loading, type of load (overhauling, etc.) and customer preference.

CAUTION

DO NOT ADJUST RLS AND ELS. THEY ARE FACTORY ADJUSTED TO PROTECT THE MU AND/OR THE DRIVEN ELEMENT AGAINST PHYSICAL DAMAGE. REFER TO POSITION/SIGNAL VARIATIONS FOR EXCEPTIONS TO FACTORY SETTINGS.

CONNECTIONS

Refer to Figures 4 and 5 for a typical interconnection between the ECU and transition box and the ECU and boiler controller. In addition to interconnecting terminal points, this drawing gives electrical contact ratings, quantity, size and type of wire, and other pertinent notations concerning terminal board jumpers, etc. Please refer to the specific

*See OPERATION (MODULATING) - MANUAL CONTROL MODE - CRANK.

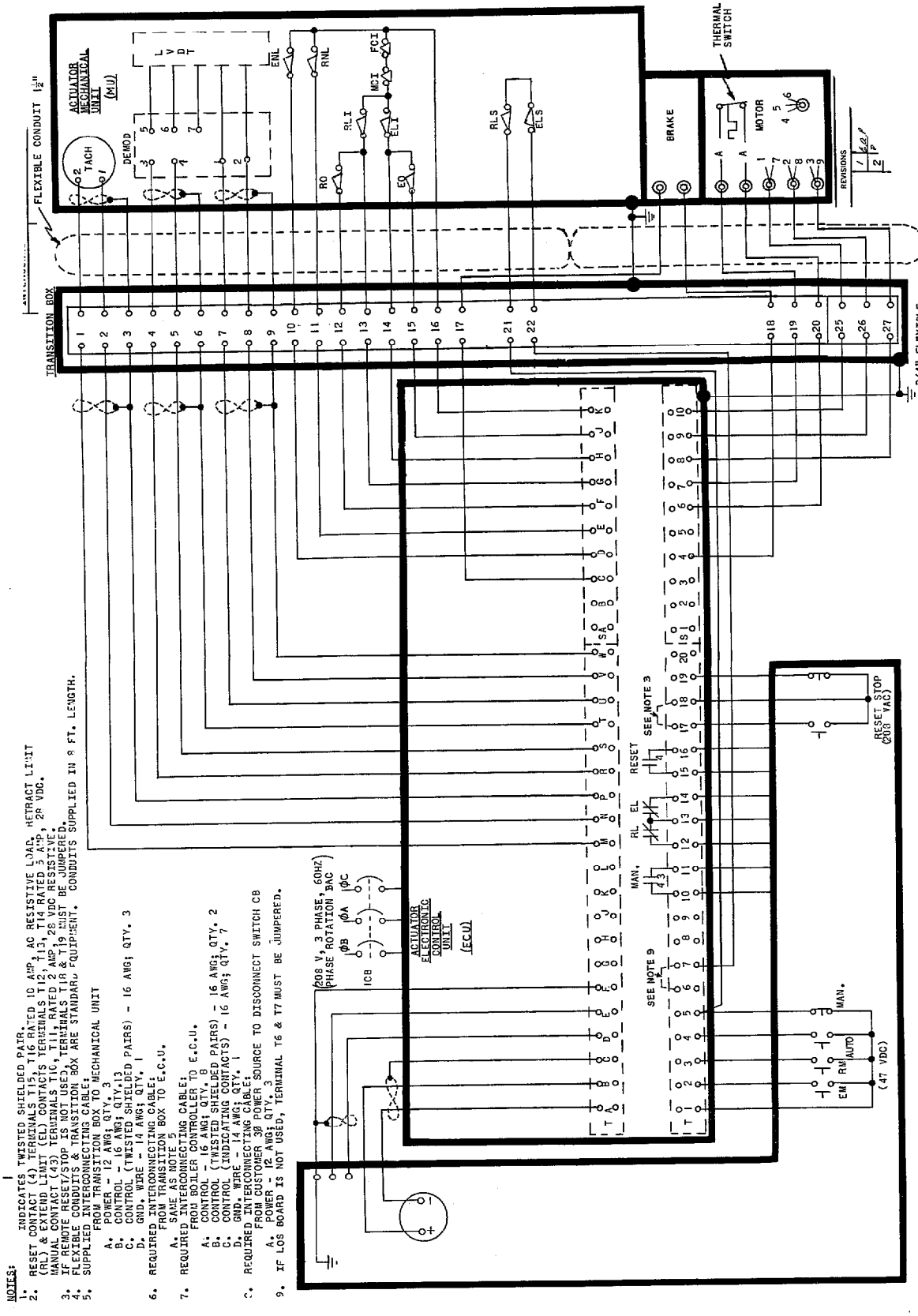


Figure 4. Actuator Interconnection (Sizes 2, 3 and 4)

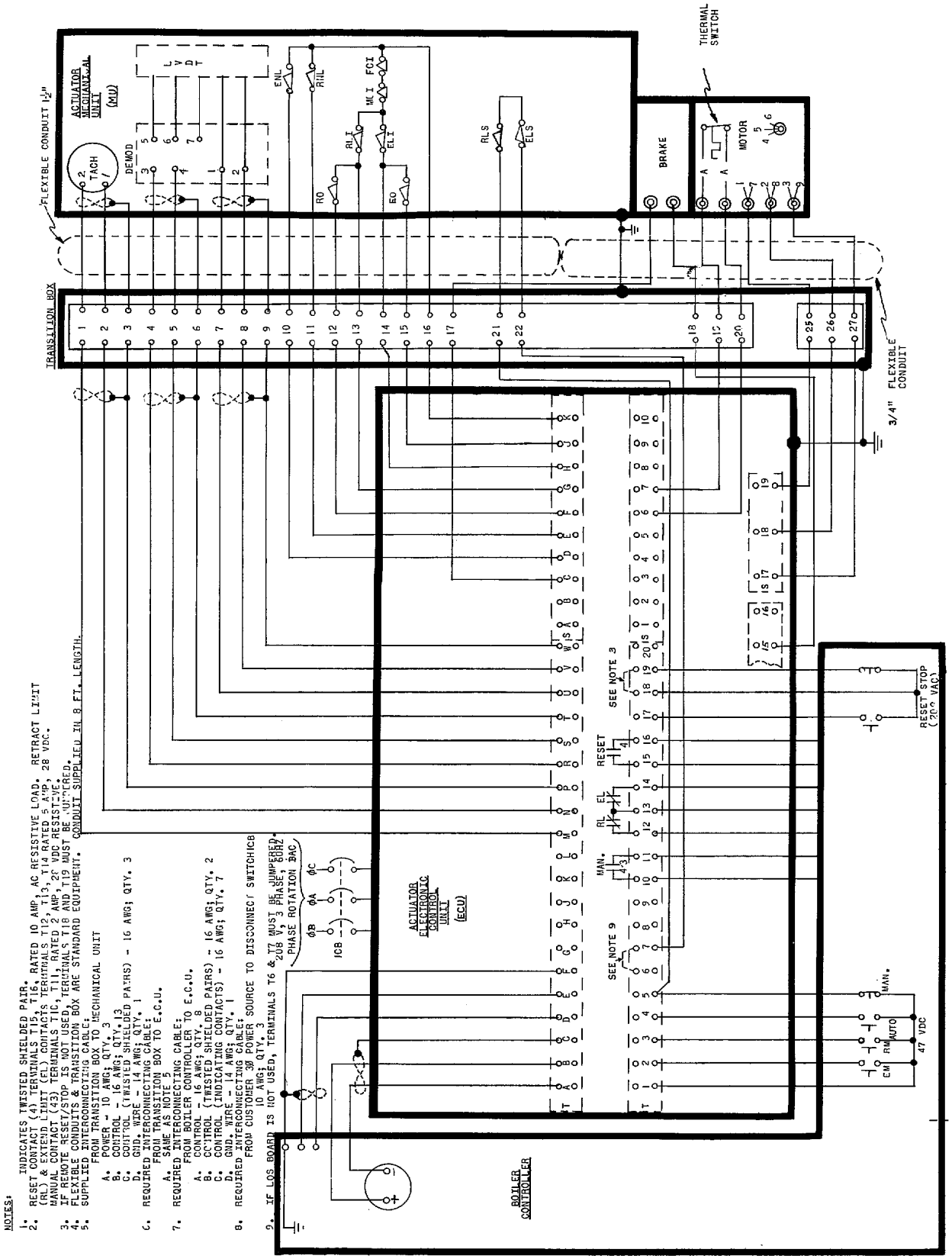


Figure 5. Actuator Interconnection (Size 5)

interconnection drawings supplied with each installation for detailed information.

OPERATION
(Modulating)

Refer to Figure 6, block drawing, or Figures 7b, 7c and 7d, elementary drawings, for the following circuit operation. Circuit numbers and termination points referred to are on the elementary drawings.

To help understand the control circuitry, let's briefly consider the events which occur in the MU and ECU. The MU consists basically of a jack and a motor. The direction and rotation of the motor turns the worm gear of the jack, moving the jack screw "in" or "out". We will convert the terms "in" and "out" to "retract" and "extend". There are a set of thyristors in phase "A" and another set in phase "B", but none in phase "C". The direction of rotation of the motor is controlled by turning "on" one set of thyristors, which supply power to the motor of a certain phase sequence. Turning this set "off" and turning another set of thyristors "on" reverses the phase sequence, reversing rotation of the motor, reversing direction of travel of the jack screw.

The ECU contains the circuitry which controls this direction as well as the speed of travel.

Circuits within the heavy lines are circuits on printed circuit cards incorporating the majority of the controlling circuitry.

AUTO CONTROL MODE

2PS (Figure 7d)

The low voltage regulated power supply for the control circuits (±15 volts, +5.1 volts and +47 volts).

2BCI (Figure 7b)

Receives a position demand signal from the boiler controller equipment. This signal is usually in the form of a current signal which is converted to a 2-10 volt signal by a loading resistor located on the incoming terminal board, terminals TD and TE, circuits 206 and 207. Polarity of TD must be +. This signal is then conditioned and "clamped" (electrical limit circuit) at ±10 volts, then fed into a summing point where it is compared with the actual position signal of the MU from the 20SC card, circuit 208. This resultant signal is amplified and becomes the position error which is the motor speed demand signal. This portion of the circuit also contains a set of limit contacts controlled by RNLS and ENLS. As the MU approaches the end of travel position, RNLS or ENLS is tripped, reducing the error signal to a low but constant value, reducing the speed of travel of the MU. The MU, therefore, will not run into the final limit at a high rate of speed.

The motor speed demand signal is then summed with a signal from a tachometer, circuit 209, to obtain a speed error signal. An absolute value (-/e/) of this speed error is derived and is summed with a deadband adjustment which controls sensitivity. The error is then amplified a final time. Also at the point where the absolute error is derived, the direction signals are developed.

In summation, the function of the 2BCI card is to develop an error signal, circuit 212, (absolute error, circuit 213) and two direction signals, circuits 211 and 210.

2TRM and MAN (Figure 7c)

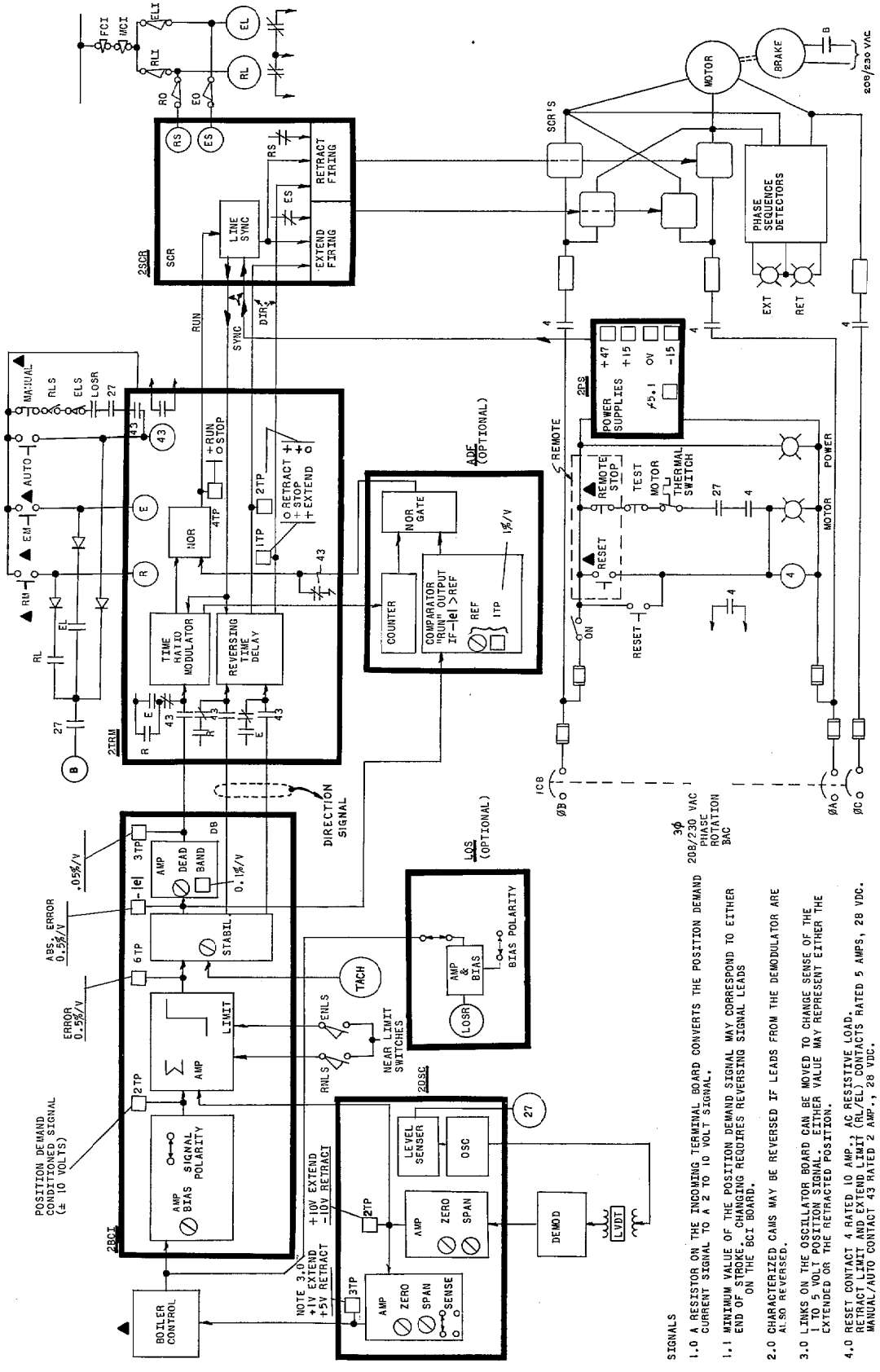
This card contains circuitry to control the mode of operation of the actuator and to synchronize and further control the error and direction signals.

Auto Mode

Depressing the "AUTO" pushbutton energizes 43 relay, sealing in through one of its own contacts, 27 relay contact, LOSR relay contact (if LOS card is used), limit switch contacts ELS and RLS, and the "MAN" (Manual) pushbutton.

All 43 relay contacts will be opposite to that shown on the drawings. It is common practice to show relay contacts in the de-energized state; in this instance 43 is shown in Manual Mode.

Circuit 212, pin 19, receives the speed error signal from the 2BCI card. The error is fed through a 43 contact to a time ratio modulator. This modulator is synchronized with the supply frequency (60 Hertz). It consists of a ramp generator that produces a sawtooth waveform which increases linearly for 10 cycles of the supply frequency, falls rapidly to zero and repeats. The speed error signal is compared with this ramp and for the time during which the error amplitude exceeds that of the ramp, a signal is generated at the TRM output which is a fractional part of the 10 cycle ramp period. If the speed error



SIGNALS

1.0 A RESISTOR ON THE INCOMING TERMINAL BOARD CONVERTS THE POSITION DEMAND CURRENT SIGNAL TO A 2 TO 10 VOLT SIGNAL.

1.1 MINIMUM VALUE OF THE POSITION DEMAND SIGNAL MAY CORRESPOND TO EITHER END OF STROKE. CHANGING REQUIRES REVERSING SIGNAL LEADS ON THE BCI BOARD.

2.0 CHARACTERIZED CAMS MAY BE REVERSED IF LEADS FROM THE DEMODULATOR ARE ALSO REVERSED.

3.0 LINKS ON THE OSCILLATOR BOARD CAN BE MOVED TO CHANGE SENSE OF THE 1 TO 5 VOLT POSITION SIGNAL. EITHER VALUE MAY REPRESENT EITHER THE EXTENDED OR THE RETRACTED POSITION.

4.0 RESET CONTACT 4 RATED 10 AMP., AC RESISTIVE LOAD. RETRACT LIMIT AND EXTEND LIMIT (RL/EL) CONTACTS RATED 5 AMPS, 28 VDC. MANUAL/AUTO CONTACT 43 RATED 2 AMP., 28 VDC.

5.0 ▲ CUSTOMER'S DEVICE

6.0 POWER REQUIREMENTS:

ACTUATOR SIZE 2: 1.3 KVA, 30, 3.3 AMPS,230 VOLTS

ACTUATOR SIZE 3 & 4: 2.5 KVA, 30, 6.2 AMPS, 230 VOLTS

ACTUATOR SIZE 5: 5.5 KVA, 30, 14 AMPS, 230 VOLTS

Figure 6. Actuator Block Diagram

PRINTED CIRCUIT BOARD LIST

	SYMBOL	TITLE	ELEM. DWG.	ORDER NO.	QTY.
	2BCI	BOILER CONTROL INTERFACE & ABSOLUTE ERROR	44C304806	44B331711-G01	1
	2ØSC	OSC - DEMODULATOR	44C304807	44B331722-G01	1
	2TRM	TIME RATIO MODULATION & MANUAL	44C304809	44B331720-G01	1
(OPTIONAL)	ADF	ADAPTIVE FILTER	44C304455	44B331712-G01	1
	2SCR	SCR FIRING & CONTROL	44C304808	44B331719-G01	1
	2PS	POWER SUPPLY	44C304149	44B331715-G01	1
(OPTIONAL)	LOS	LOSS OF SIGNAL	44C305892	44B337311-G01	1

CUSTOMER NOTES:

1.) ⏚ INDICATES ZERO REFERENCE POTENTIAL, NOT CASE GROUND.

2.) THE LOCATION OF THE PRINTED CIRCUIT BOARDS IS SHOWN BY THE NUMBERS APPEARING NEAR THE SQUARE WHICH REPRESENTS THE P.C. BOARD. EXAMPLE: 0104 WOULD BE IN THE 4TH SLOT IN THE P.C. BOARD RACK.

3.) INDICATES SHIELDED CABLE.

4.) INDICATES TWISTED WIRES.

5.) ◎ INDICATES CUSTOMER CONNECTION TERMINAL POINTS.

6.) SOLID LINE INDICATES PRINTED CIRCUIT BOARDS.

7.) □ TEST POINT.

8.) ⊠ TEST POINT (FOR FACTORY USE).

9.) ✳ INTERNAL ADJUSTMENT.

10.) ▲ CUSTOMERS DEVICE.

11.) ■ DEVICES IN ACTUATOR MECHANICAL UNIT.

Figure 7a. A. C. Actuator Modulating Control Master Elementary Diagram

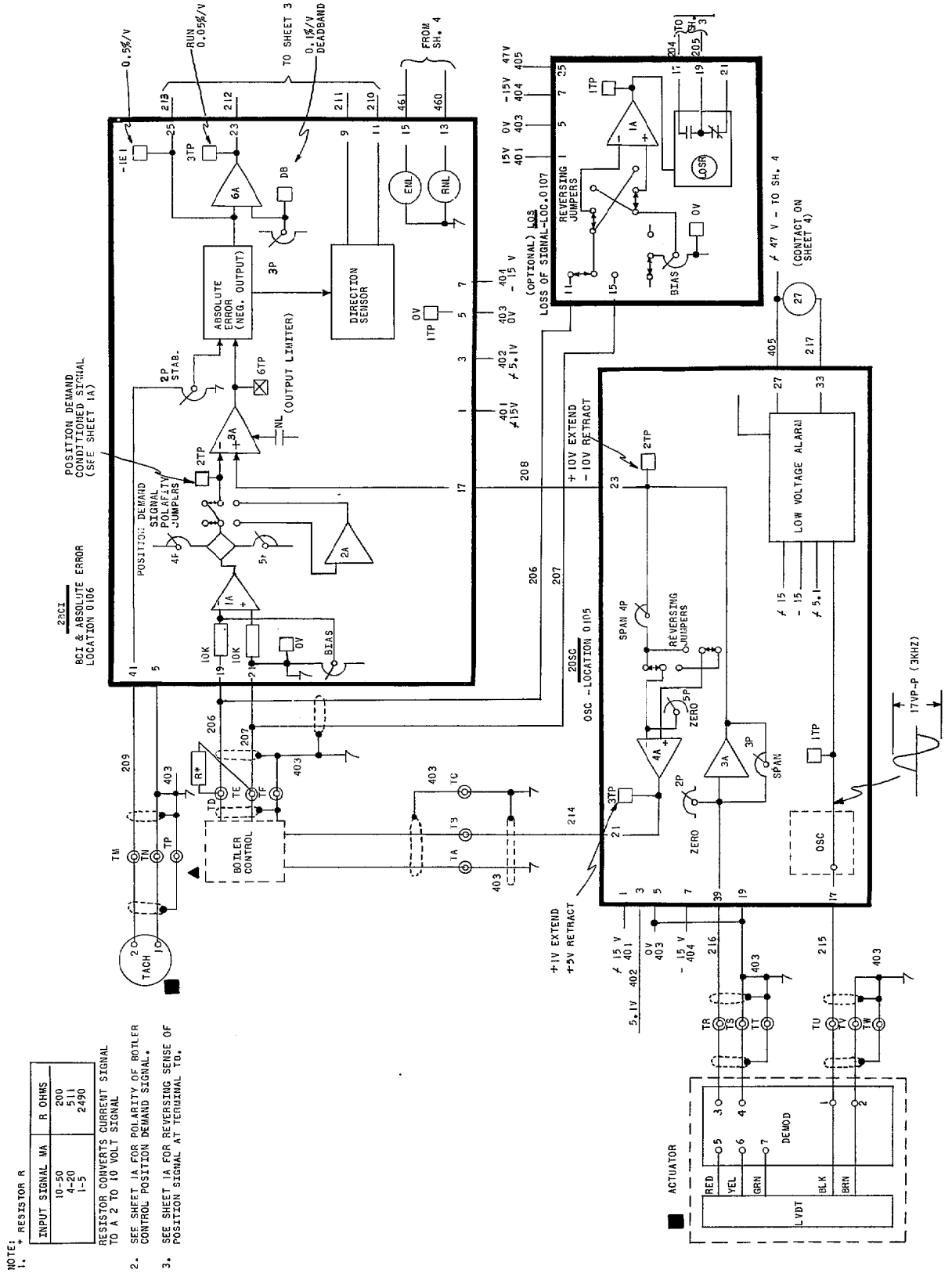


Figure 7b. A. C. Actuator Modulating Control
Master Elementary Diagram

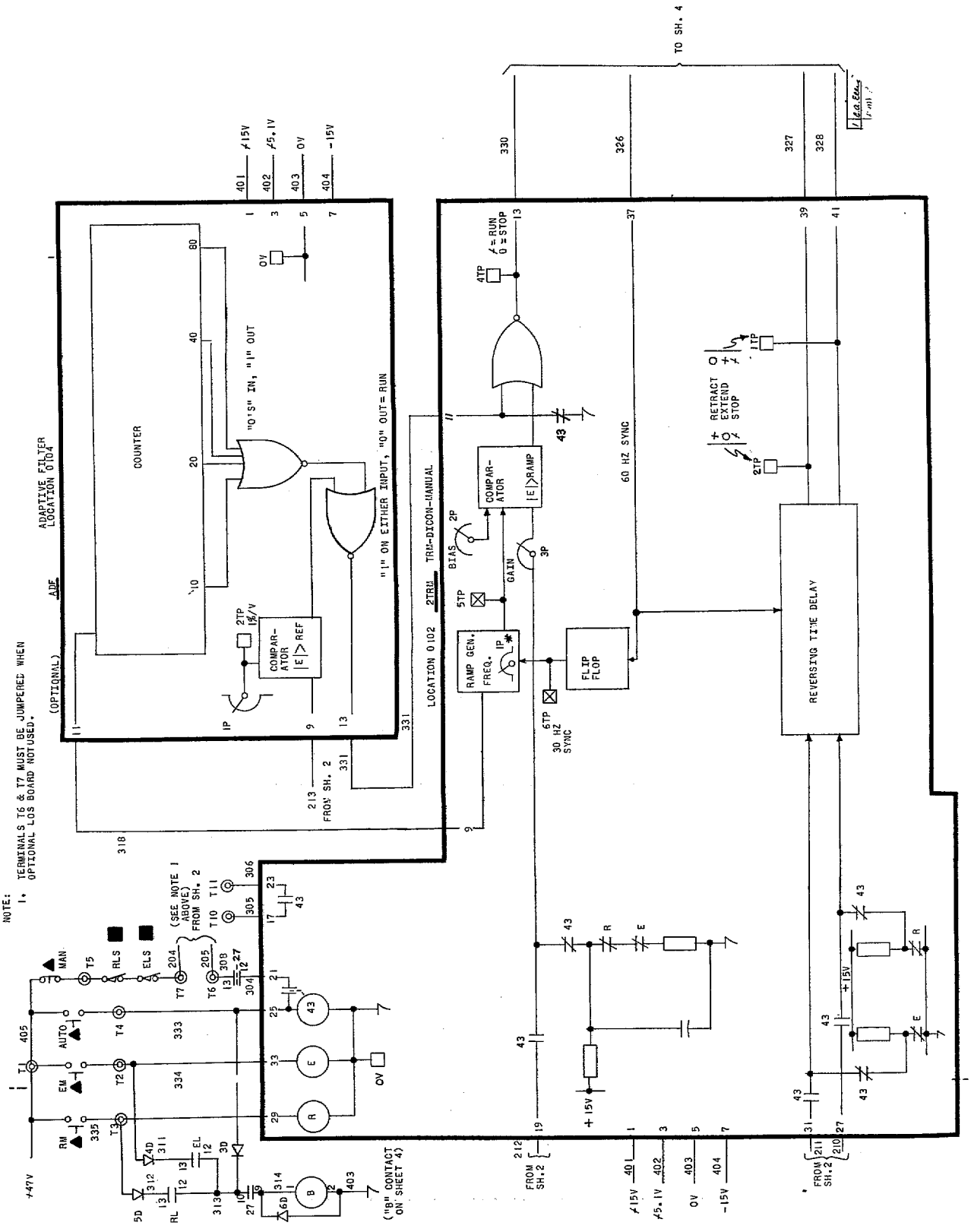


Figure 7c. A. C. Actuator Modulating Control Master Elementary Diagram

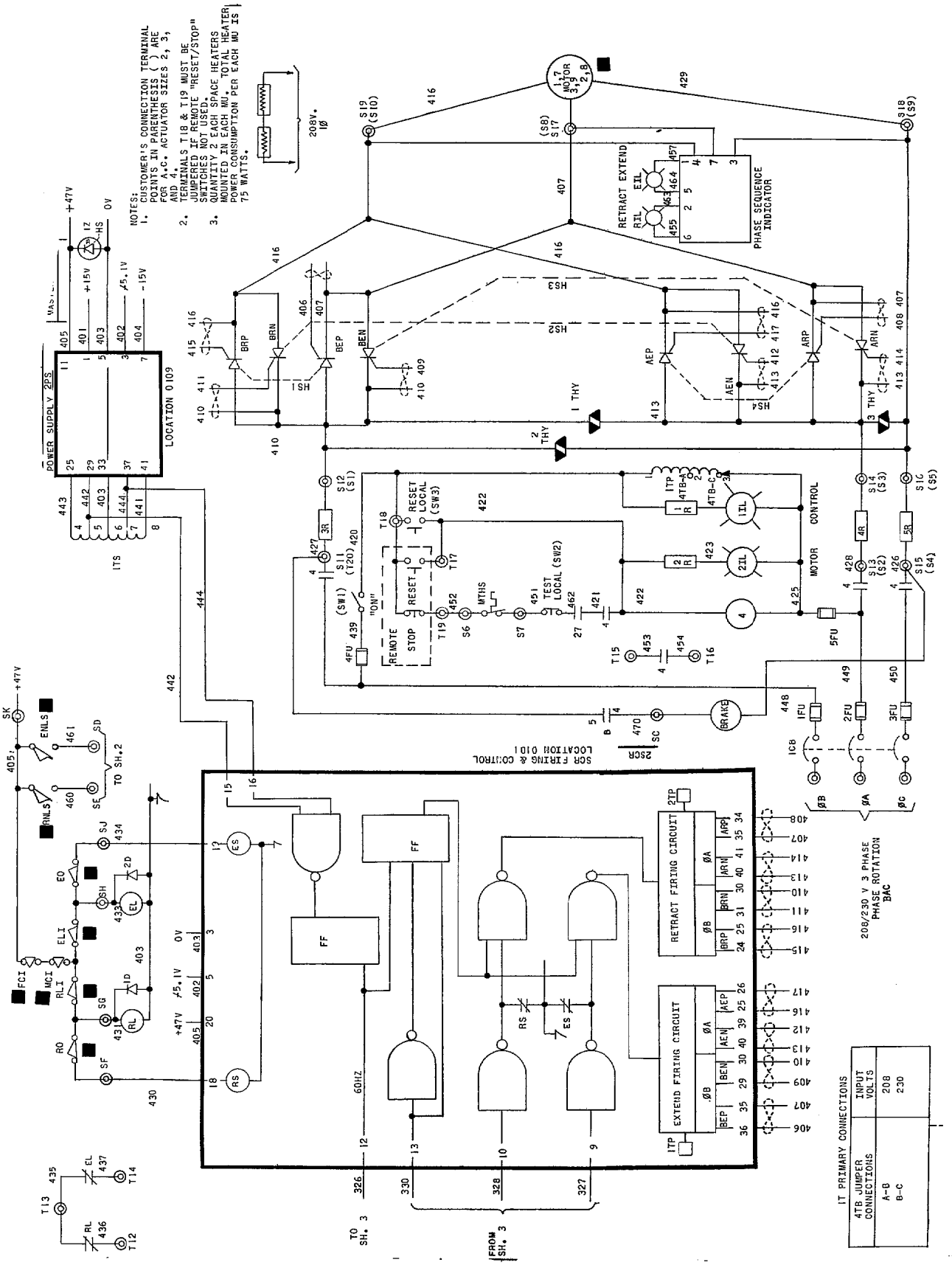


Figure 7d. A. C. Actuator Modulating Control Master Elementary Diagram

signal is of a slightly higher amplitude, this will cause the output pulse of the TRM to increase in duration. The duration of the output pulse is therefore proportional to the amplitude of the speed error.

The output of the TRM is then fed into a "NOR" logic. Out of the "NOR" logic is a "RUN" and "STOP" signal, the "RUN" when the output is at a "1" and "STOP" when at a "0".

Circuits 211 and 210, pins 31 and 27, receive the direction signals from the 2BCI card. These signals are fed through 43 contacts to a reversing time delay circuit. This circuit inhibits reversing of the direction signals for one cycle of the 60 Hertz supply frequency. This permits the set of conducting thyristors to turn "off" before turning "on" the other set of thyristors, preventing a line-to-line short circuit. Test points 1TP and 2TP are the conditioned direction signals. This circuit, like the ramp generator, is synchronized with the supply frequency.

Manual Mode circuitry will be discussed under "Manual Control Mode".

Note that when in Auto Mode, relay B is always energized, releasing the brake.

2SCR (Figure 7d)

This printed circuit card combines the time ratio modulated run signal, circuit 330, pin 13, and the conditioned direction signals, circuit 328, pin 10, and circuit 327, pin 9, from the 2TRM card with a line frequency synchronizing signal. The synchronizer utilizes a combination of flip-flop and gate circuitry for selective and positive firing of the "sets" of thyristors through the extend and retract firing circuits. It controls the application of the firing signals in integral numbers of cycles to prevent half-wave operation of the motor.

The synchronizer also provides the synchronizing signal, circuit 326, pin 12, for the reversing time delay and the ramp generator on the 2TRM card.

Relays ES (extend stop) and RS (retract stop) inhibit thyristor firing as determined by limit switch action. They are normally energized and will inhibit thyristor firing when de-energized.

The resultant function of this board therefore is to combine the "run" and direction signals from the 2TRM card and synchronize them with the line frequency and limit switch conditions to inhibit or permit the firing of the thyristors.

20SC (Figure 7b)

This card contains a 3 kilohertz oscillator which supplies excitation, circuit 215, pin 17, to the primary of an LVDT in the position feedback assembly/circuitry mounted in the MU. The oscillator incorporates a voltage regulator and temperature compensation circuits to maintain the output at a constant amplitude and frequency. The output can be varied from approximately 3 to 8 volts rms. It is factory set and should be maintained at 6 volts rms. Once set, increasing or decreasing the amplitude will increase or decrease the position feedback signal, altering the limit of MU travel. 3P must be adjusted so the output at 2TP will be + or -10 volts. Variations to this setting will be discussed under "Position Signal Variations".

The secondary of the LVDT is fed into a demodulator, converting it to a DC level. (Details of this assembly/circuitry will be discussed under "Position Feedback Assembly/Circuitry". Briefly, the purpose of this assembly/circuitry is to develop a signal which is proportional to the MU position.) The output of the demodulator is fed back to pin 39, circuit 216, of this card where it is electrically balanced (2P) and amplified (gain potentiometer 3P).

The output of 3A or 2TP, circuit 208, pin 23, is adjusted for +10 volts and -10 volts, representing extend and retract MU positions. This is then fed to pin 17 of the 2BCI card where it is compared with the position demand signal of +10 volts and -10 volts.

The output at 2TP also feeds a circuit which conditions the feedback signal to a +1 volt to +5 volt signal, circuit 214, 3TP, pin 21, for customer use indicating actuator position. +1 volt or +5 volts may represent either the extend or retract position as determined by the placement of the jumpers.

Location 1:	+ 5 volts = retract; +1 volt = extend
Location 2:	+ 5 volts = extend; +1 volt = retract

Potentiometers 4P and 5P are adjustments for zero and span. Jumpers and potentiometers are factory set for +1 volt = extend and +5 volts = retract.

This card also contains circuitry which monitors the supply voltages of +15, -15, +5.1 and 47 and the 3 KHz oscillator. If any of these voltages should happen to fail, relay 27 will be de-energized, transferring the

system to Manual Mode, applying the brake and removing the system from electrical control. The MU will remain in the last set position.

The previous circuits on PC cards 2BCI, 2TRM, 23CR and 20SC were intentionally described in specific sequence to follow the flow of the input position demand signal from the boiler controller equipment to the controlling direction and speed signals of the actuator when operating in Auto Mode.

OPTIONAL CIRCUITRY

ADF (Figure 7c)

The high level performance achieved by modern boiler controllers combined with signals which are often noisy will result in far more minor corrections of an actuator's position than are actually required.

This board has been designed specifically to eliminate these unnecessary corrections and thus extend the life of the mechanical parts several fold. This circuitry, at the same time, will not degrade the overall high level performance inherent in the boiler controller.

The position error is compared with an adjustable reference. If the position error exceeds the reference, a signal is passed through a logical NOR gate to another NOR gate, permitting a "run" signal. Hysteresis incorporated in the comparator requires the error be reduced to 0.5% to reset the comparator to the "no-run" condition.

A four-bit binary counter triggered from the output of the 10 cycle ramp generator on the 2TRM board has outputs of 10, 20, 40 and 80 cycles duration. These are combined to obtain an output of 10 cycles for every 160 cycles (2.67 seconds) which feeds the logical NOR gate to permit the motor to run and correct small errors. Thus, for all noisy signals of low magnitude, the motor is permitted to run only 1/16 of the time.

The ADF connector is wired in on all ECU panels, although the customer may not prefer its use. If the customer requires this optional feature at a later date, this card can be installed and operating within a few moments. Just insert the card into the slot marked "ADF" and adjust the reference point.

LOS (Figure 7b)

The circuitry on this card senses the level of the position demand signal at the incoming terminal board, terminals TD and TE. If the amplitude falls below a predetermined level, relay LOSR is de-energized transferring the mode of operation to Manual control and applying the brake.

The object of this circuit is to have the MU remain in the last set position or travel very little from the last set position when a loss of the position demand signal occurs. This is possible because the response of the electronics to the failure is much faster than the response of the mechanical parts to the failure.

Without this circuit the MU can travel to one or the other of the end limits. The exact end limit will depend on the placement of jumpers on the 2BCI card.

Like the ADF card, an LOS connector is also wired in on all ECU panels. Remove jumpers from terminals T6 and T7 and insert the card into the slot marked LOS. 1P is then set to the desired value.

MANUAL CONTROL MODE - ELECTRICAL (Figure 7c)

Depressing the MAN momentary pushbutton de-energizes 43 relay. The contacts are as shown on the block and elementary drawings.

The speed error and direction signals from the 2BCI card no longer have control. Their functions have been replaced by pushbuttons "RM" (Retract - Manual) and "EM" (Extend - Manual).

Depressing EM energizes relay "E", opening a contact, allowing a capacitor in an RC timing circuit to charge. This charging rate is applied to the time ratio modulator where it is compared with the ramp circuit. The resultant output at 4TP is a series of pulses with variable durations. The duration increases with time until approximately 2 seconds, when the capacitor is almost completely charged. The output will then be at a constant + (plus) level indicating full speed "run" condition. This action permits the operator to jog the MU accurately to a new position.

Another "E" contact opens, applying a signal to the input of the reversing time delay circuit, causing the output at 2TP to be at a "0" (extend) and 1TP to be at a "1" (stop).

Depressing RM energizes relay "R", producing the same circuit action of the RUN/STOP signal as the "E" relay. However, the direction signals are reversed, i.e., 2TP will be at a "1" (stop) and 1TP at a "0" (retract).

If both pushbuttons are accidentally depressed, 1TP and 2TP will be at a "1" (stop) and no action will occur.

Relay "B" is energized only when either RM or EM is depressed, otherwise it is de-energized, applying the brake.

Diodes 4D and 5D prevent interaction of control relays R and E when RM or EM is depressed or when in Auto mode to prevent interaction between 43 and RM and EM. Diode 3D also prevents any interaction between RM and EM and 43.

It may be noted that minimum circuitry is required in Manual mode. The operation of the circuitry following the output of the 2TRM card is the same as the Auto mode of control. Besides the 2PS card, 2TRM and 2SCR are the only circuitry cards required for electrical Manual control, providing reliable control. Circuitry on cards 2BCI and ADF are removed from control by the 43 relay. The LOS circuitry is eliminated because its output contact from the LOSR relay is in the energizing/de-energizing 43 relay circuit. The only function provided by the 20SC card in addition to the level sensing circuit, is the feedback position signal fed back to the boiler controller. This position signal may or may not be required, depending upon the application of the actuator. Other end control element signals may be used to determine the actuator's position.

MANUAL CONTROL MODE - CRANK

This mode is used when the actuator cannot be operated electrically by either of the other two modes, such as a power failure. This mode may also be used in the initial installation when mounting and adjusting linkages, resetting limit switch cams or repositioning the LVDT.

Removing the hand crank from the right side cover of the MU will inhibit the thyristor firing circuits by opening MCI (Manual Crank Interlock) switch. On actuator sizes 2, 3, and 4, remove the brake access cover (the correct cover is indicated by an arrow on the brake nameplate) to manually release the brake. Remove the plug (GE nameplate) from the end of the motor and insert the crank. The MU will move 1" for every 25 full turns of the crank.

The size 5 actuator uses a standard ratchet wrench with a 1/2" female drive. The conversion socket in the end of the motor shaft is exposed but with a protective guard. The brake release lever is also exposed on the side of the brake. Turning the lever 90° will release the brake. The MU will travel 1" for every 16 full turns of the crank.

Turning all cranks clockwise as viewed from the motor end will extend the MU.

POSITION FEEDBACK ASSEMBLY/CIRCUITRY (Figures 1A, 1B, 7b and 8)

This assembly consists of an LVDT, cam follower arm, cam and a demodulator. The cam is mounted on the assembly which moves as the MU extends and retracts. The cam follower arm follows this movement, changing position of the LVDT core. With the LVDT primary excited by the 3KHz oscillator on the 20SC card, the LVDT secondary is connected so there will be no output when the core is electrically centered. These secondary windings are connected to the demodulator, terminals 5, 6, and 7 converting the AC to a DC level. As the core moves toward winding "A" more voltage is induced in that winding with terminal 3 being more positive than terminal 4. With the core nearer winding "B", more voltage is induced in this winding and less induced in winding "A", terminal 4 now becoming more positive than terminal 3. Terminals 3 and 4 therefore are the actuator's actual position signal, which is fed back to the ECU to be amplified and conditioned.

The use of an LVDT provides a highly accurate and linear device with infinite mechanical life.

FEEDBACK CAM (Figures 1B and 9)

All MU's are shipped with a linear cam. An additional cam will be supplied for characterizing at the job site. Square and square root curves and increments of 0.15 inch horizontal lines and 1 inch vertical lines are silk-screened onto the rectangular cam as guides for custom characterizing as shown in Figure 9.

When characterizing a cam, the characterization should equal the full cam rise of 1 1/2".

CAUTION

THE RATE OF CHANGE OF ANY CONTOURING MUST NOT EXCEED A 16:1 RATIO. A LARGER RATIO WILL CAUSE UNSTABLE OPERATION.

The position signal amplifier 3A on the 20SC card must be readjusted to compensate for any rise less than 1 1/2". Refer to "Position Signal Variations" for determining the appropriate setting.

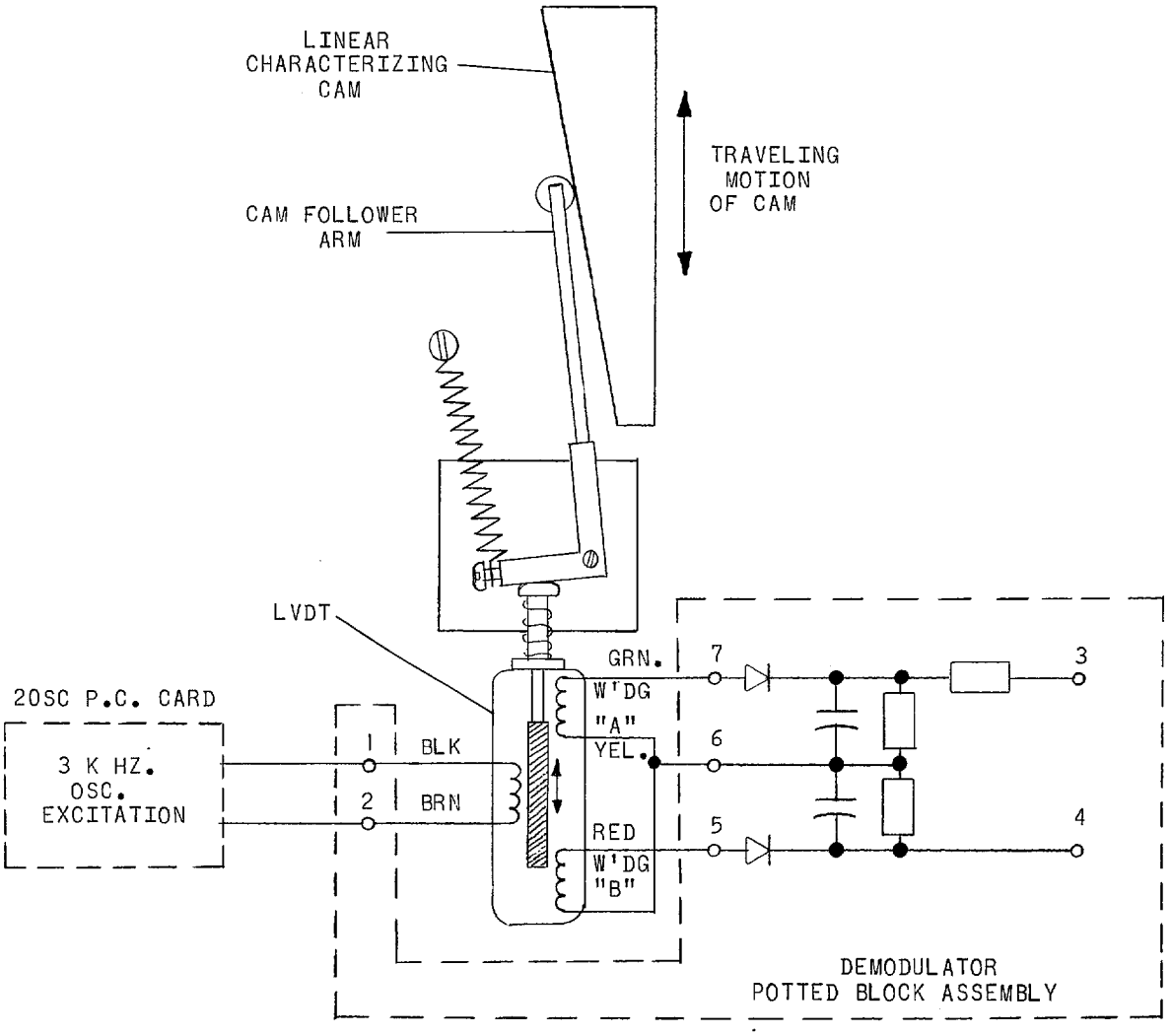


Figure 8. Position Feedback Circuitry

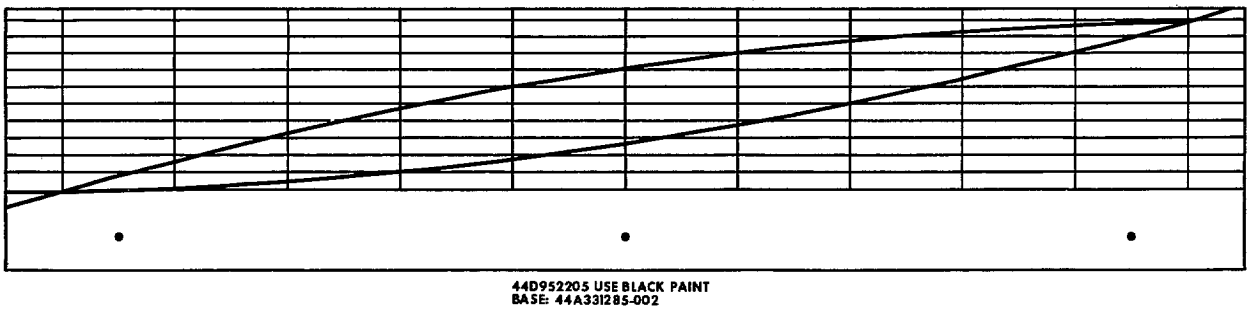


Figure 9. Characterizing Cam

A particular application may require a cam to be reversed. Reversing demodulator output leads 3 and 4 will cause the actuator to respond accordingly.

BRAKE (Figure 1B)

All brakes are single phase AC operated disc brakes. They consist basically of a disc, two pressure plates and a solenoid. With the solenoid de-energized, springs apply a force to one pressure plate compressing the disc between the two pressure plates. Energizing the solenoid releases the force on the plate allowing the disc to rotate freely with the shaft.

The brake is a double "C" face type mounted between the motor and the MU housing. The driving shaft from the motor is directly coupled to the brake shaft with a coupling hub. The opposite end of the brake shaft is coupled to the jack shaft with a 3-jaw coupling.

Brakes for sizes 2, 3 and 4 have a manual release lever inside the brake housing. An arrow on the nameplate indicates which one of two access covers to remove to release the brake.

Brakes for a size 5 have the manual release lever externally.

CAUTION

DO NOT OPERATE ACTUATOR WITH BRAKE IN MANUAL RELEASE POSITION AND WIRES DISCONNECTED OR NO POWER APPLIED TO SOLENOID. MANUAL RELEASE IS INTENDED ONLY FOR EMERGENCY AND INITIAL STARTUP, NOT AS A SUBSTITUTE FOR FULL ELECTRICAL RELEASE.

In the Auto mode the solenoid is energized, releasing the brake. The brake is not required because of the low efficiency jack and the closed loop system operation.

In Manual mode, depressing either RM or EM pushbutton releases the brake, conversely, releasing either pushbutton immediately applies the brake. This provides the operator with precise control of the driven element.

The brake also assists the jack, immediately stopping the MU when a power failure, loss of low voltage supply or oscillator, or loss of demand signal (when LOS card is used) occurs. A failure in the limit-of-travel circuit or the electrical limit circuit may cause the MU to overtravel too far into the extreme limits, damaging the MU and/or the driven element. RLS and ELS will prevent this by transferring the system to Manual and immediately applying the brake, minimizing overtravel.

MOTOR (Figure 1B)

All motors are 3-phase, 60 Hz, AC induction with high slip design D (GE type KR) characteristics. They are totally enclosed, non-ventilated with NEMA "C" face mounting. A self-resetting, normally closed thermo switch provides overload protection.

The 1 HP motor (size 2 actuator) and the 2 HP motor (size 3 and 4 actuators) have a special shaft with a 3/8 inch Allen hex nut recessed in one end for insertion of the manual crank. The 5 HP motor has a special shaft extension onto which an adapter is mounted for manual crank control.

OVERFORCE LIMITER ASSEMBLY (Figure 1A)

This device is mounted between the jack screw flange and the output pipe flange. It mechanically limits excessive forces, protecting the MU and/or the driven element against physical damage.

Within the housing are disc springs stacked and preset to approximately 120% of maximum rating. If the unit encounters an overforce in either direction, the disc springs begin to collapse, tripping limit switch EO (Extend Overforce) or RO (Retract Overforce), removing power from the motor. The springs will reset when the excessive force is removed, permitting normal operation. These springs are factory set and can not be reset at the job site. However, they can be set at specific lower values prior to testing and shipment by notifying the factory of a special application.

JACK (Figure 1A)

The jack is the "muscle" part of the actuator system. It must move the load smoothly and efficiently, retaining high accuracy while constantly reversing direction of travel. Its low efficiency gearing means the motor is energized only when moving to a new position. This also means cooler operation contributing to long life.

The excellent mechanical accuracy of the actuator system is due to an anti-backlash feature. When the jack screw is under compression load, the bottom of its thread surfaces are supported by the top thread surfaces of the worm gear. The anti-backlash nut is adjusted downward by the shell cap, forcing the nut's bottom thread surfaces against the upper thread surfaces of the screw. Thus, backlash between the worm gear threads and the jack screw threads is reduced to a minimum. This minimum must not be less than .002 inches. A closer tolerance will increase wear rapidly.

CAUTION

THE LIMIT OF TRAVEL IN INCHES PER HOUR MUST BE TAKEN INTO CONSIDERATION FOR EACH APPLICATION. THE FOLLOWING VALUES ARE FOR 100% ACTUATOR LOAD AT 25°C AMBIENT.

SIZES 2, 3, AND 4 NOT TO EXCEED 400 INCHES/HOUR. SIZE 5 NOT TO EXCEED 240 INCHES/HOUR.

EXCESSIVE HEAT BUILD-UP OCCURS WHEN THESE OPERATING RATES ARE EXCEEDED, REDUCING THE LUBRICATING PROPERTY OF THE GREASE. IN ADDITION, THE EXPANSION OF METAL PARTS TAKES PLACE INCREASING FRICTION, INCREASING HEAT BUILD-UP, CREATING A RUNAWAY CONDITION TO WHERE THE METAL PARTS SEIZE, DAMAGING THE JACK PERMANENTLY.

SPECIAL LUBRICANT SHELL DARINA #2 E.P. GREASE OR ITS EQUIVALENT MUST BE USED TO OBTAIN THE ABOVE MAXIMUM CAPABILITIES.

PHASE SEQUENCE INDICATOR (PSI)

This circuit determines the direction of travel of the MU by detecting the phase voltage differences. The indicating median are two neon lamps mounted on the front panel of each ECU. When lit the lamp denotes the function described above it.

ADJUSTMENTS, TEST AND INITIAL OPERATION

PRELIMINARY

1. Install input loading resistor on terminals TD and TE for the following current signals.

10-50 ma signal; 200 ohms
4-20 ma signal; 511 ohms
1-5 ma signal; 2490 ohms

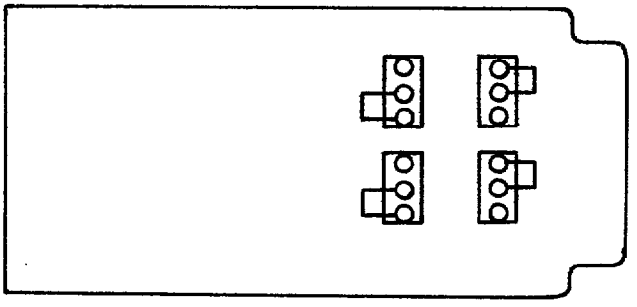
No resistor is required if the input is a 2-10 volt signal.

2. Remove 2BCI card and check location of jumpers.

a. With jumpers in location 1, minimum current signal at TD-TE will retract the actuator.

b. With jumpers in location 2, minimum current will extend the actuator.

3. Remove LOS card (if used) and check location of jumpers per sketch. This will not be necessary on later design cards where jumpers are eliminated.



4. Remove 20SC card and check location of jumpers.

a. With jumpers in location 1, actuator position indication is:

+1 volt represents extend
+5 volt represents retract

b. With jumpers in location 2, actuator position indication is:

+1 volt represents retract
+5 volt represents extend

5. When the LOS card is not used, terminal T6 must be jumpered to terminal T7. Without this jumper, the Auto mode of control will be inoperable.

6. When the remote stop and reset pushbuttons are not used, terminal T18 must be jumpered to T19. Without the jumper, relay 4 will not seal in, thus no power will be supplied to the motor.

7. Check the input voltage connection to the primary of 1T transformer. Mounted above the transformer on the right side panel of the ECU is a 3-point terminal board, 4TB. Place the jumper on terminals A-B for 208 volts input supply or on terminals B-C for 230 supply volts.

For any supply voltage greater than 230 volts, a 3-phase step-down power supply transformer with a 230 volt secondary must be used and the jumper placed on terminals B-C for 230 volt, 1-phase operation for the actuator control circuits. Refer to Figure 6 for total power consumption required for each actuator system.

CAUTION

CORRECT PHASING OF THE INPUT SUPPLY VOLTAGE IS VERY IMPORTANT. INCORRECT PHASING WILL RESULT IN THE ACTUATOR TRAVELLING IN THE OPPOSITE DIRECTION TO THAT COMMANDED BY THE INPUT DEMAND SIGNAL. THERE WILL BE NO END LIMIT TRAVEL PROTECTION. BE SURE THE CONNECTION TO THE 3-PHASE DISCONNECT SWITCH ADJACENT TO EACH ECU ARE OF THE PROPER PHASE SEQUENCE AS LABELED ABOVE EACH SWITCH CONNECTION.

CIRCUIT CHECKS AND ADJUSTMENTS

When making the following checks, be sure the MU is not positioned on any limits, unless otherwise directed.

When at the limits, the most accurate circuit adjustments will be obtained when the limit switches have just been depressed and a "click" heard.

Because the electrical Manual mode does not incorporate the near end-of-travel limits, RNLS and ENLS, it should not be used for the following checks and adjustments. The combination of the closed loop servo system (Auto mode) and the near end-of-travel limits produces the ultimate accurate settings. The electrical Manual mode may allow the MU to overtravel a small amount (depending on the type of load), producing erroneous voltage levels. If the pertinent PC card settings are adjusted using these levels, the system accuracy, when in Auto mode, pertaining to limits of travel, will be affected in some instances more so than others. Use the Manual crank.

All voltage measurements should be taken with a digital voltmeter. All factory settings are made with a digital voltmeter capable of 3 decimal point readings.

Turn the ECU "on" but do not depress the RESET pushbutton.

ADF P.C. CARD

Wiper of 1P, test point 2TP, is factory set at 1 volt. This is 1% of the input demand signal voltage span. This will be adequate for most applications. However, if the demand signal is excessively noisy, increase 2TP voltage to a value which will eliminate these unnecessary corrections of the MU but still retain the desired accuracy. This should be done after all other circuit checks and adjustments are made. Remove this card from its connector and leave in this position until all PC card checks and adjustments are made.

2BCI P.C. CARD (Card Extender Required)

Adjust 3P for desired deadband (sensitivity) at test point DB. For every volt at DB, the input demand signal can change ±4 millivolts without any change in MU position. DB is factory set at 5 volts.

With jumpers in location 1 and the input demand signal at minimum (1, 4 or 10 ma. or 2 volts) check for +10 volts at 2TP. Adjust 4P if necessary. With the input demand signal at maximum (5, 20 or 50 ma. or 10 volts) check for –10 volts at 2TP. Adjust 5P if necessary. Alternate input demand signals from minimum to maximum and repeat 4P and 5P adjustments.

20SC P.C. CARD

Check for a 17 volt peak-to-peak, 3KHz, ±300 Hz sine wave at 1TP, waveform (1). Rarely will it be necessary to readjust 1P for the correct amplitude.

See Caution on following page.

CAUTION

ANY ALTERATION IN AMPLITUDE WILL AFFECT THE RANGE OF MU TRAVEL. LIMIT SWITCHES ARE FACTORY SET WITH THE OSCILLATOR OUTPUT AT 17 VOLTS P-P (6 VOLTS RMS).

Although the oscillator frequency is a contributing factor to the level of the position feedback signal, it does not affect the amplitude of this signal (2TP) nearly as much as the setting of 1P. The ±10% variation in the frequency is not as important as the fact that whatever the frequency is within this tolerance, it must remain at this frequency within a much closer tolerance to prevent a level change in the position feedback signal.

Check the demodulator gain and balance amplifier 3A. With the MU at the retract limit, 2TP should be -10 volts. Conversely, when at the extend limit, 2TP should be +10 volts. The exception to this is when the position cam in the MU is reversed, requiring terminals 3 and 4 of the demodulator to be reversed, thus 2TP being reversed. Adjust 3P if necessary, to obtain + and -10 volts at the limits. Refer to "POSITION SIGNAL VARIATIONS" for adjustment of 2P.

With customer position feedback signal (terminal TA, TB and TC) jumpers in the desired location (PRELIMINARY, Item 4), check the voltage levels, 3TP, at the retract and extend limits. If necessary, adjust as follows:

a. With a jumper, short 2TP to "0" volts.
b. Adjust 5P for 3 volts at 3TP.
c. Remove jumper and position the MU to one of the limits.
d. Adjust 4P for either +1 volt or +5 volts, depending on jumper locations. When at the opposite limit, 3TP will be +5 volts or +1 volt, the opposite end of the span voltage.

2TRM P.C. CARD (Card Extender Required)

Using a dual-channel oscilloscope, place channel 1 probe on 6TP waveform (3) and channel 2 probe on 5TP waveform (2). Adjust the oscilloscope horizontal sweep frequency so 5 square waves at 5TP are displayed on the screen. Superimpose the ramp waveform of 6TP on the square wave. If necessary, adjust 1P for a 5:1 ratio, i.e., one sawtooth ramp for every 5 square waves. This synchronizes the ramp generator with the line frequency.

Remove 2BCI card from its connector. With the scope probe on 4TP, waveform (4), check for positive going pulses. If the pulses exist, adjust 2P clockwise so the pulses just disappear. Turn 2P clockwise approximately 5° more.



Energize 43 relay. This may be done by placing the actuator system in Auto mode. Do not depress the RESET pushbutton. Apply +10 volts to pin 19. With the oscilloscope probe on 4TP, check for negative going pulses. If the pulses exist, adjust 3P counterclockwise so the pulses just disappear. Turn 3P counterclockwise approximately 5° more.



Return the actuator system to Manual mode. Insert the 2BCI card into its connector.

Check 1TP and 2TP for a "1" (+5 volt) or a "0" (zero volt) signal by depressing RM and EM. Refer to Figure 7c for the direction signal table.

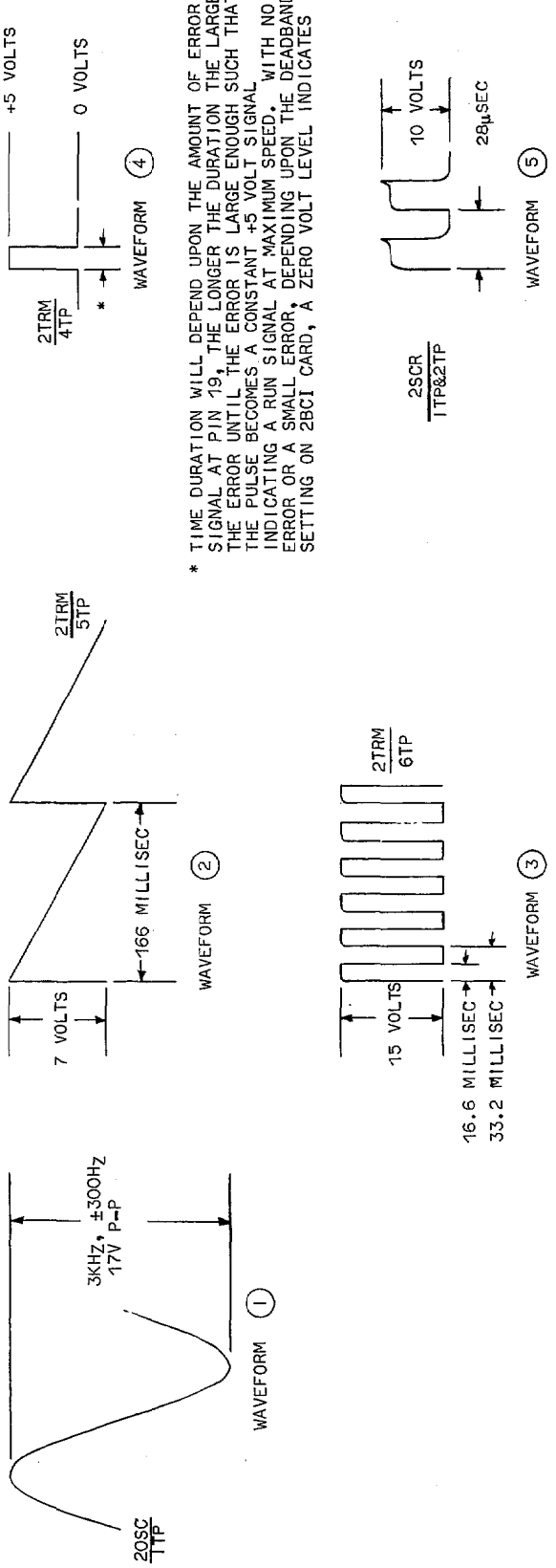
2SCR P.C. CARD

With the oscilloscope probe on 1TP and the system in Manual mode, depress EM pushbutton. This is the extend firing circuit oscillator, waveform (5).

With the oscilloscope probe on 2TP, depress RM pushbutton. This is the retract firing circuit oscillator, waveform (5).

NOTE

When either pushbutton is depressed, the waveform will pulse, then remain constant. This action was described under MANUAL


+5 VOLTS
0 VOLTS
2TRM / 4TP
*
WAVEFORM (4)
* TIME DURATION WILL DEPEND UPON THE AMOUNT OF ERROR SIGNAL AT PIN 19, THE LONGER THE DURATION THE LARGER THE ERROR UNTIL THE ERROR IS LARGE ENOUGH SUCH THAT THE PULSE BECOMES A CONSTANT +5 VOLT SIGNAL INDICATING A RUN SIGNAL AT MAXIMUM SPEED. WITH NO ERROR OR A SMALL ERROR, DEPENDING UPON THE DEADBAND SETTING ON 2BCI CARD, A ZERO VOLT LEVEL INDICATES
2SCR / 1TP&2TP
10 VOLTS
28μSEC
WAVEFORM (5)
2TRM / 5TP
7 VOLTS
166 MILLISEC
WAVEFORM (2)
2TRM / 6TP
15 VOLTS
16.6 MILLISEC
33.2 MILLISEC
WAVEFORM (3)
2OSC / 1TP
3KHz, ±300Hz
17V P-P
WAVEFORM (1)

WAVEFORMS

NOTE (Continued)

CONTROL MODE - ELECTRICAL, allowing the operator to jog the MU accurately to a new position. This applies pulses of power to the motor for approximately the first two seconds, then full power for the remainder of the time the pushbutton is depressed.

LOS P.C. CARD

Adjust 1P to the desired value per the following example.

EXAMPLE: Loading resistor (terminals TD-TE) = 200 ohms.
Loss-of-signal value = 2 ma.
Then IR = E or .002 x 200 = .4 volts.
Set 1P wiper at .4 volts.
If the current through the loading resistor falls below 2 ma, relay LOSR will be de-energized.

PLACING THE ACTUATOR SYSTEM IN AUTO CONTROL

1. Set the position demand signal to mid-value.

2. Turn the ECU "ON/OFF" switch to the "ON" position.

3. Depressing "RESET" pushbutton, the MU should travel to mid-position (when using the linear feedback cam).

4. Adjust 2P on 2BCI card to stabilize the system. Only a fine adjustment of 2P should be required. It is factory adjusted with an average setting derived from various loading conditions. Changing the position demand signal in a step function to other values will provide easier setting of 2P.

5. Set ENLS and RNLS cams so the MU will travel into the final limit position in approximately five steps.

POSITION/SIGNAL VARIATIONS

1. When the length of travel of the MU is less than the maximum capability, make the following adjustments, observing first a few precautions.

a. Adjust only the extend limit cams ELI, ENLS and ELS.

b. Do not adjust RLS cam.

c. Adjust ELI cam for the desired limit of travel. This new maximum limit must not be set less than 40% below the maximum capable limit. Consult factory for limits of travel beyond this. Position the MU midway between zero inches and the new maximum. Loosen the two LVDT mounting screws and move the LVDT to the new electrical zero position while observing the output of the demodulator, terminals 3 and 4, with the digital voltmeter. Position the LVDT for a voltage reading as close to zero volts as possible and tighten the two screws. NOTE: After tightening these screws, the output voltage will no longer be at zero. Again loosen the screws and change LVDT positions very slightly. After a few tries, a "feel" will be developed as to the distance required to set the LVDT for zero volts output after tightening the screws.

Position the MU to one of the maximum limits. This may be done in the electrical MANUAL mode, but not allowing the MU to travel into the RLI or ELI limit. Traveling into the limits should be done with the manual crank only, providing maximum accuracy when in AUTO mode. Note the voltage reading at 2TP of the 20SC card. Position the MU at the other limit and note the voltage reading. Adjust 2P on the 20SC card until the voltages are equal but opposite in magnitude.

Adjust 3P for 10 volts (polarity will be determined by connections to terminals 3 and 4 of the demodulator potted block assembly) with the MU at one limit. Position the MU at the other limit. 2TP should be reading 10 volts with the opposite polarity. Alternate position limits and adjustments until + and -10 volts is obtained.

Adjust ENLS as previously discussed.

Adjust ELS as follows:

Position the MU so ELI switch roller is midway on ELI cam. Adjust ELS cam until a "click" is heard from the ELS switch.

2. When a replacement card is installed, the previous checks and adjustments discussed for that particular card should be performed.

MAINTENANCE

ELECTRONIC CONTROL UNIT

The ECU requires no routine maintenance.

MECHANICAL UNIT

1. Jack - When the jack requires lubrication, use only Shell Darina #2 E.P. grease or its equivalent. How often lubrication is required will depend upon several factors.

   a. Travel in inches/hour.
   b. Ambient temperature.
   c. Loading.

The frequency of lubrication, therefore, may range from once every 2 to 3 months to once every 1 to 2 years.

Like lubrication, the frequency of the backlash setting will depend upon the same factors. Initial readjustment may occur as early as once the first six months, but may not be required again for several years. To minimize backlash, loosen the two set screws on shell cap (where the drive screw protrudes from the jack housing) and tighten down on this cap until the desired amount of backlash is obtained, then retighten the set screws. Do not adjust backlash to less than .002".

2. Brake

   a. Sizes 2, 3, and 4 (Figures 10, 12)

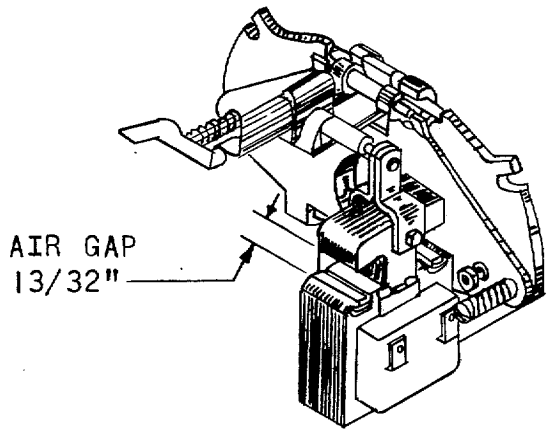


Figure 10. Brake Solenoid

Normal disc wear will cause the solenoid lever to move upward, increasing solenoid air gap and stopping time. When the gap reaches approximately 11/16", the brake must be adjusted. This gap is measured between the solenoid frame and the solenoid plunger. This may be visually inspected by removing the access covers. To adjust for lining wear, turn both wear adjustment screws equal amounts in a clockwise direction until 13/32" air gap is reached.

For brakes mounted in a vertical position (MU mounted horizontally), check the air gap as follows. Hold plunger in a free horizontal position and move it toward the solenoid frame until spring pressure is felt. Measure the air gap and adjust to 13/32". After adjusting, release and check again.

b. Size 5

This brake incorporates an automatic adjustment for disc wear.

c. Removal (Sizes 2, 3, and 4)

Remove left side cover (as viewed from front) from MU.

Disconnect the conduit and electrical leads from brake to motor.

Loosen two screws and remove the magnet assembly.

Remove the four nuts from the studs that hold the motor/brake assembly to the actuator frame and remove the motor/brake assembly.

Note and mark position, then remove the detent wheel/coupling half. Mark position on new brake shaft.

Separate and remove, noting the position, that section of the brake housing which contains the shaft. Do not at any time depress the solenoid plunger which would release and misalign the disc(s).

Note the position of the remaining section of the brake housing, then remove, leaving the brake hub-coupling and four mounting studs.

Loosen the two Allen set screws and remove the hub-coupling.

d. Installation (Sizes 2, 3, and 4)

Remove hub from the new brake, using caution to preserve alignment of the disc(s). Be sure the stabilizing springs are positioned in their proper location after hub is removed.

With the key in place on the motor shaft, slide the hub (square end first) onto the shaft to within 7/16" of the motor mounting face. Tighten the two set screws.

Position properly and slide the end plate of the brake housing (with disc(s)) carefully onto the four mounting studs. Rotate the motor shaft to align and engage the hub-coupling into the disc(s) and seat the brake on the motor "C" face.

Remove the access covers, position this section of housing properly, and slide it onto the mounting studs, rotating the brake shaft to engage into the hub.

Mount the detent wheel;coupling half. Tighten screws.

Make conduit and electrical connections.

Mount the motor/brake assembly onto the actuator frame, carefully mating the brake/jack coupling. Align coupling halves as closely to center as possible for noise and vibration free operation.

Mount the magnet assembly and make adjustments to align the detent wheel and magnet assembly per Figure 13.

e. Removal (Size 5)

Remove left side cover (as viewed from front) from the MU.

Disconnect the conduit and electrical leads from brake to motor.

Loosen bolts in both halves of the brake/jack coupling. Slip the "rubber tire" out of the brake half of coupling.

Remove the two bolts holding the brace to the feet of the motor.

Remove the four bolts holding the motor/brake assembly and remove this assembly.

Remove brake housing bolts. Note position, then separate brake housing from end plate.

Depress the solenoid plunger and place a screwdriver through the neck of the lever arm, across the cast opening in the support plate, to lock the brake mechanism in the manual release position.

Remove the entire support assembly, pressure plate and friction disc.

Remove end plate from motor.

Remove brake hub-coupling from motor shaft.

f. Installation (Size 5)

Remove hub-coupling from new brake. With key in place on motor shaft, slide hub-coupling (square end first) onto motor shaft to within 1" of motor "C" face. Tighten the set screws.

Disassemble the new brake per previous disassembly instructions of old brake.

Mount end plate to motor.

Reassemble friction disc, stationary disc and pressure plate in correct sequence and position.

Mount support plate assembly, tightening screws alternately. Remove screwdriver.

Depress the solenoid plunger manually and allow it to snap into position. Repeat several times to set air gap on solenoid.

Assemble housing to end plate, rotating the shaft to engage key into hub keyway.

Tighten housing bolts to 118 ft. lbs.

Make conduit and electrical connections.

Mount motor/brake assembly but do not fully tighten.

Snap the "rubber tire" into the coupling, being sure it is completely seated.

Align the motor/brake shaft and the jack shaft as closely as possible for noise and vibration free operation.

Fully tighten the four motor/brake assembly bolts.

Manually release the brake. Rotate the shaft and check shaft and coupling alignment. Fully tighten coupling bolts.

Actuator systems in Auto mode will operate for several years without any need for disc wear adjustment.

3. Motor

Removal and Installation - Follow the same procedure as the removal and installation of the brake.

OPERATION

(Non-Modulating)

Refer to Figures 11 and 12 for the following description and circuit operation.

This type of control, also called "ON-OFF", requires minimum circuitry.

The actuator consists of a Controller Panel Unit (CPU) and a MU.

CPU

Contains a 3-phase circuit breaker, step-down transformer and a reversing controller.

Step-down transformer 1T supplies 48 volts AC to energize the relay coils.

The reversing controller transfers power and interchanges phasing to the motor.

It may contain one or two complete control circuits, therefore, one panel is capable of controlling two MU's.

MU

The MU is basically the same as the modulating units with the following items removed.

Tachometer
LVDT
Cam Follower
Characterizing Cam
Demodulator Assembly
Extreme end of travel limit
switches and associated cams

OPERATION (Refer to 44C305713)

Three-phase power is fed through the circuit breaker to a set of contacts on the "E" relay and to a set on the "R" relay. When "E" is energized, contacts close applying power of a certain phase sequence to the motor, driving the MU to the extend limit. When "R" is energized, power is applied to the motor of another phase sequence driving the MU to the retract limit. This control, like the modulating type, is phase sensitive. Incorrect phasing will drive the MU the opposite direction to that which was commanded, damaging the MU and/or the driven element.

Each coil is energized through a set of series contacts, switches and pushbuttons. Contact MTHS and switches FCI, MCI, EO and RO provide same functions as in the modulating unit. ELS and RLS are the extend and retract limit switches. "EXT" and "RET" are the customer's direction control contacts. EH and RH contacts are controlled by two overload heaters in series with two phases of the motor. ELI and RLI are the extend and retract limit switch contacts for customer use. Positioning of ELI and RLI cams is usually identical with ELS and RLS.

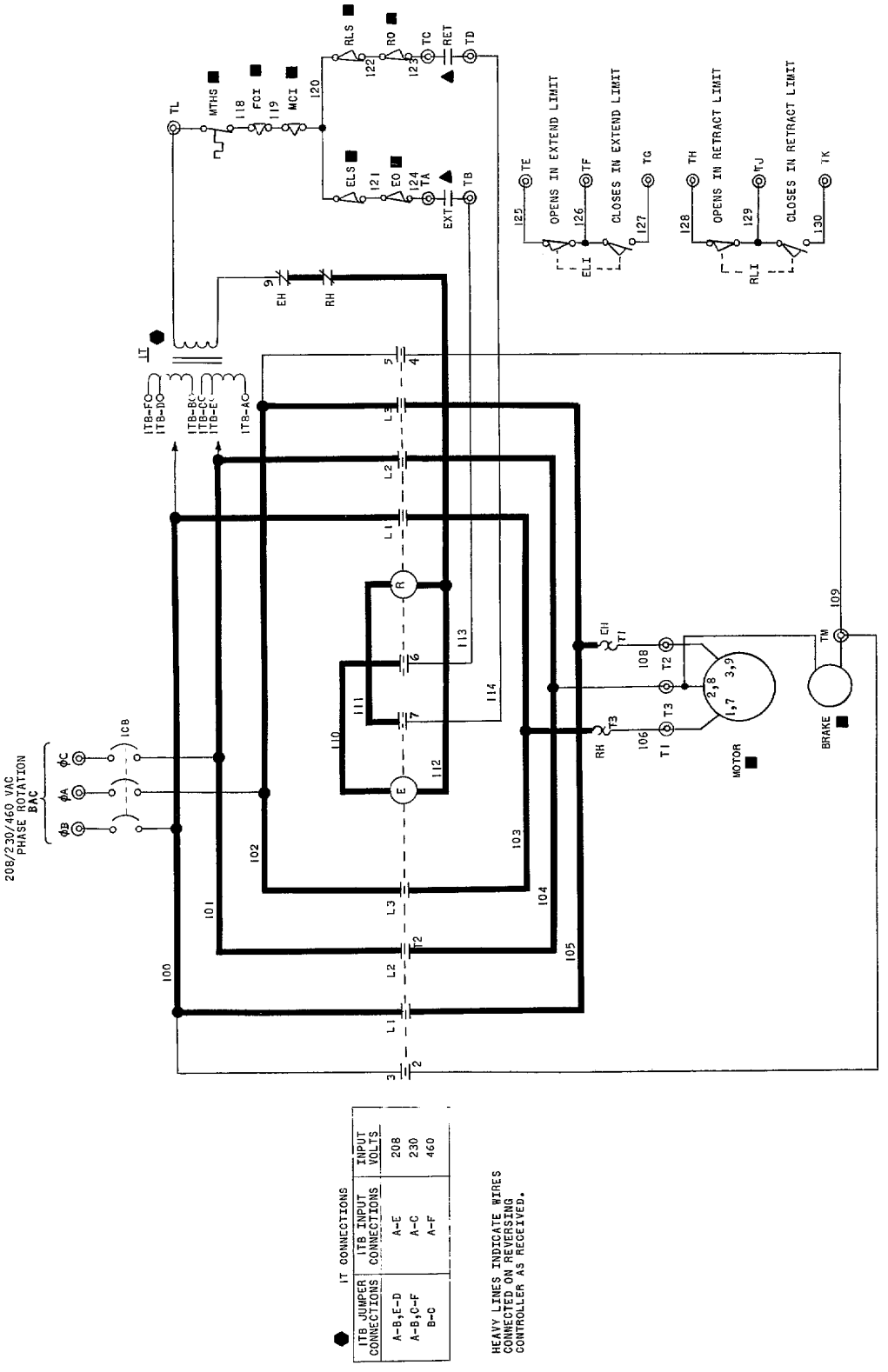


Figure 11. A. C. Actuator Non-Modulating Control Elementary

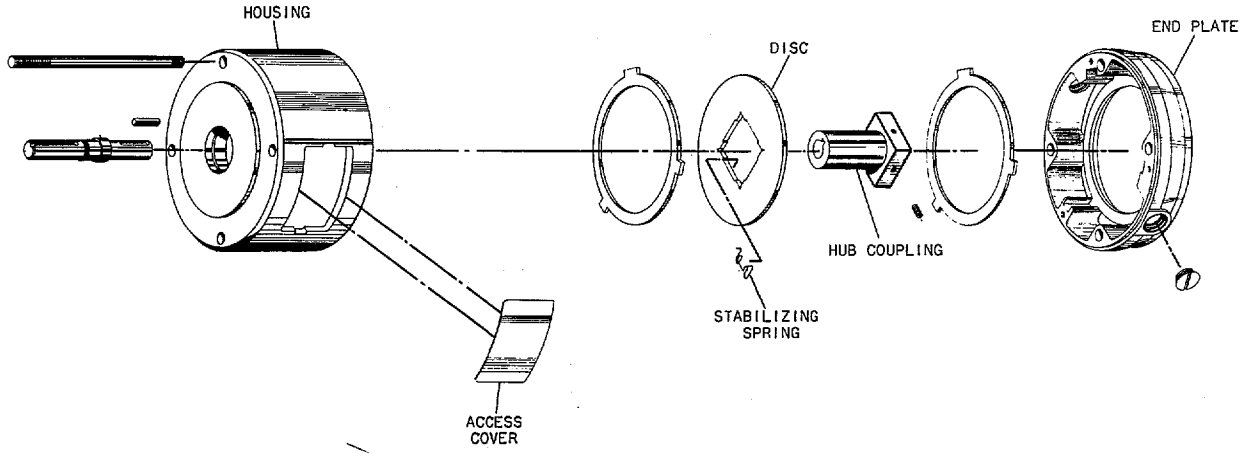


Figure 12. Brake (Sizes 2, 3, 4)

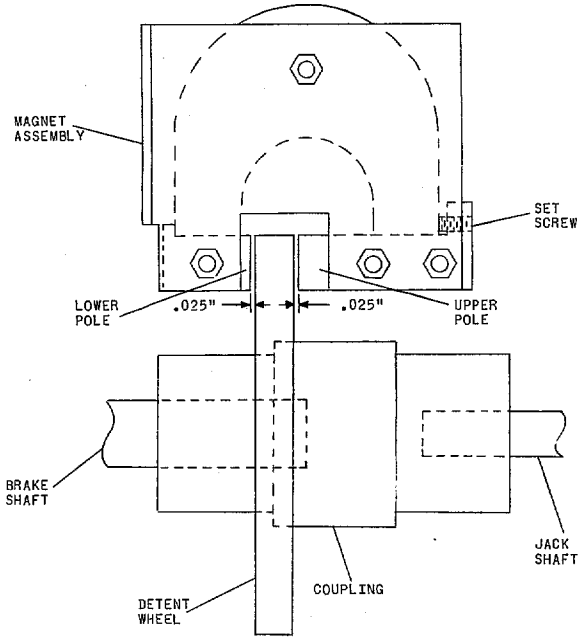


Figure 13. Magnetic Detent

TROUBLESHOOTING

TROUBLE	POSSIBLE CAUSE
1. MU will not travel in Manual mode	a. Blown fuse(s) - high or low current b. 3-phase circuit breaker open c. Front cover of MU not in position d. Mechanical crank not in storage position e. Faulty "4" relay f. Faulty "B" relay g. Faulty motor h. Faulty brake i. Faulty 2PS card j. Faulty 2SCR card k. Faulty 2TRM card
2. MU will not travel in Auto mode	a. All of trouble #1 above b. Faulty 2BCI card c. Faulty 20SC card d. Loss of position demand signal (when LOS card used)
3. MU will travel only in one direction	a. Faulty SCRs b. Faulty 2TRM card c. Faulty 2SCR card d. MU in overforce state
4. Inadequate travel	a. Improper limit cam settings b. LVDT improperly adjusted c. Faulty demodulator assembly d. Incorrect loading resistor e. Check for proper minimum and maximum demand signal f. Improper setting of 2P and/or 3P - 20SC card g. Faulty 20SC card h. Improper setting of 4P and/or 5P - 2BCI card i. Faulty 2BCI card
5. MU travels to a limit and remains there	a. Loss of demand signal (when LOS card not used) b. Faulty LVDT c. Faulty demodulator d. Faulty 20SC card e. Faulty 2BCI card
6. MU travels into limit(s) at high rate of speed (may also trip from Auto to Manual mode)	a. ENLS and RNLS cams improperly set b. Faulty 2BCI card

(continued)

TROUBLESHOOTING (Continued)

TROUBLE	POSSIBLE CAUSE
7. MU begins to travel but stops quickly	a. MU in overforce state (check excessive friction in linkages or the driven element) b. Faulty "B" relay c. Faulty brake
8. Motor hums (60 Hz) but will not rotate	a. Blown high current fuse in one phase b. Faulty SCR(s) c. MU in overforce state d. Faulty brake e. Faulty motor
9. Unstable operation	a. P.C. cards not inserted properly into their connectors b. Improper setting of 2P - 2BCI card c. Faulty 2BCI card d. Broken tachometer belt e. Faulty tachometer
10. Will not transfer from Manual mode to Auto mode	a. MU positioned at an extreme limit (RLS or ELS) b. Faulty 2TRM card c. Faulty LOS card d. Faulty 20SC card e. Faulty 2PS card f. Faulty "27" relay
11. Will not reset	a. Motor thermal switch activated due to excessive heating. Check for cause. b. Blown low current fuse(s) c. Faulty 20SC card[d. Faulty 2PS card e. Faulty "4" relay f. Faulty "27" relay
12, Noisy operation	a. Check jack lubricant. Be sure to use proper lubricant - see text. b. Backlash may be excessive. Refer to MAINTENANCE - Jack, for adjustment. c. Check guide rods. A thin coating of a light oil is all that is required. d. Check clearance of the detent wheel/magnet assembly

RENEWAL PARTS

When ordering renewal parts, the following information should be given:

1. Catalog number stamped on the part, with complete description including use and location.

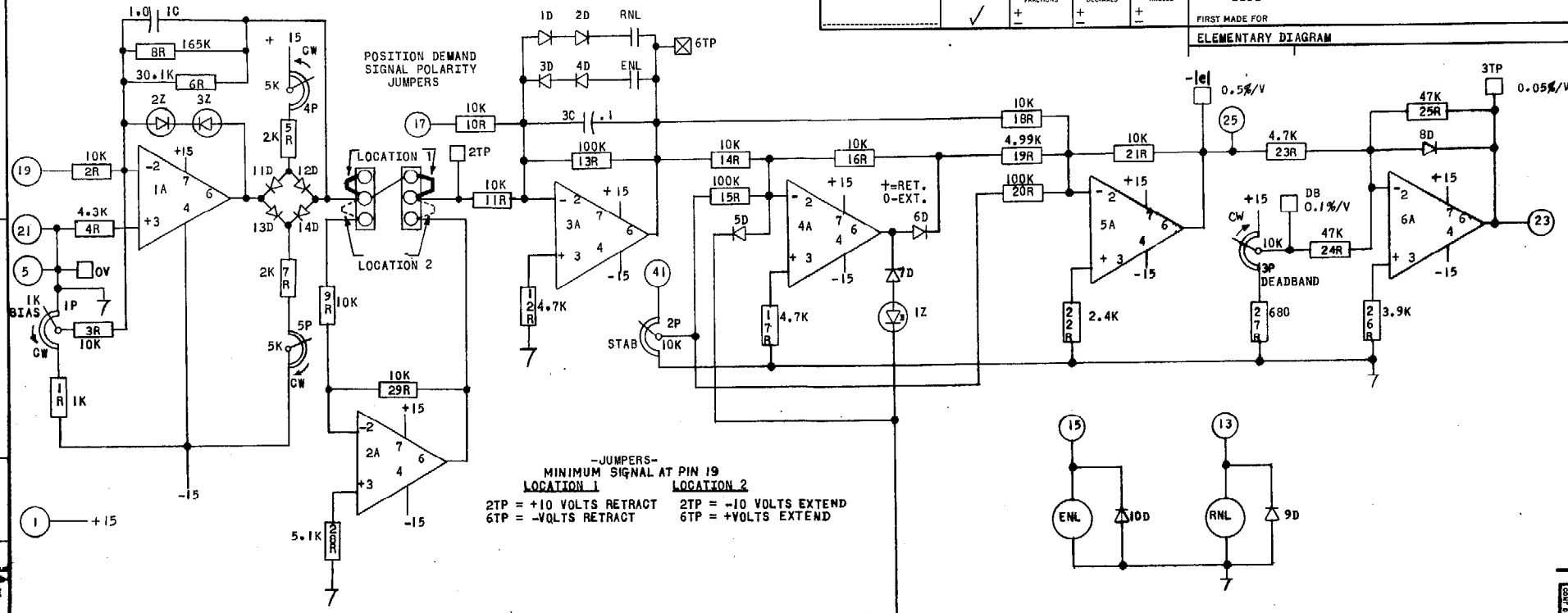
2. Complete nameplate data appearing on the assembly that included the component.

3. If possible, data on original order on which equipment was first supplied, including all numerical references.

GENERAL ELECTRIC

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
2BCI
FIRST MADE FOR
ELEMENTARY DIAGRAM



□ = TEST POINT ON EDGE OF BOARD.
⊠ = TEST POINT NEAR COMPONENT.

EXT. ADJ.:
- 1P - ZERO VOLTS
- 2P - TACHOMETER STABILITY
- 3P - DEADBAND
- 4P - +10 VOLT LIMIT ADJUST
- 5P - -10 VOLT LIMIT ADJUST

TEST POINTS:
- 0V - ZERO VOLTS
- 2TP - CONDITIONED POSITION DEMAND SIGNAL
- 3TP - ERROR (+)
- -|e| - ABSOLUTE ERROR
- DB - DEADBAND
- 6TP - ERROR (+ OR -)

1R	933B590F10
2,3,9,10,11, 14,16,18,21,29R	128A4057G10
4R	933B590F43
5,7R	128A4057F20
6R	128A4057G301
8R	128A4057H165
12,17,23R	933B590F47
13R	128A4057H10
15,20R	933B590H10
19R	128A4057F499
22R	933B590F24
24,25R	933B590G47
26R	933B590F39
27R	933B590E68
28R	933B590F51
1P	44B318618F10
2,3P	44B318618G10
4,5P	44A333423F50
1C	933B597J10
3C	933B597H10
1D-8D,11D-14D	44B232028-001
9,10D	44B232019-004
1Z	1N753A
2,3Z	1N962B
1A-6A	44A332168-001
1,2G	44A333402-002
RNL,ENL	3SAV5004M1
TERM. BD.	44B310014-201
U-CLIP	44A219567-001
TEST POINT	44B412208-001

REVISIONS

1	E.A. Clem	June 6, 1971
2	R.M. Bulesky	Nov. 9, 1971
3	E.A. Clem	June 28, 1972

PRINTS TO: 11

MADE BY E. Clem 6/27/72
APPROVALS MBC
DRAWN C. Lotts NOV. 19, 1971
CONTROL & CONT. DEVICES DIV OR DEPT WAYNESBORO, VA. LOCATION
44C304806 CONT ON SHEET SH NO

GENERAL ELECTRIC

44B331711 FL 1

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

APPLIED PRACTICES	SURFACES	FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

PRINTED WIRING BOARD

2BC1 (F.C.F.)

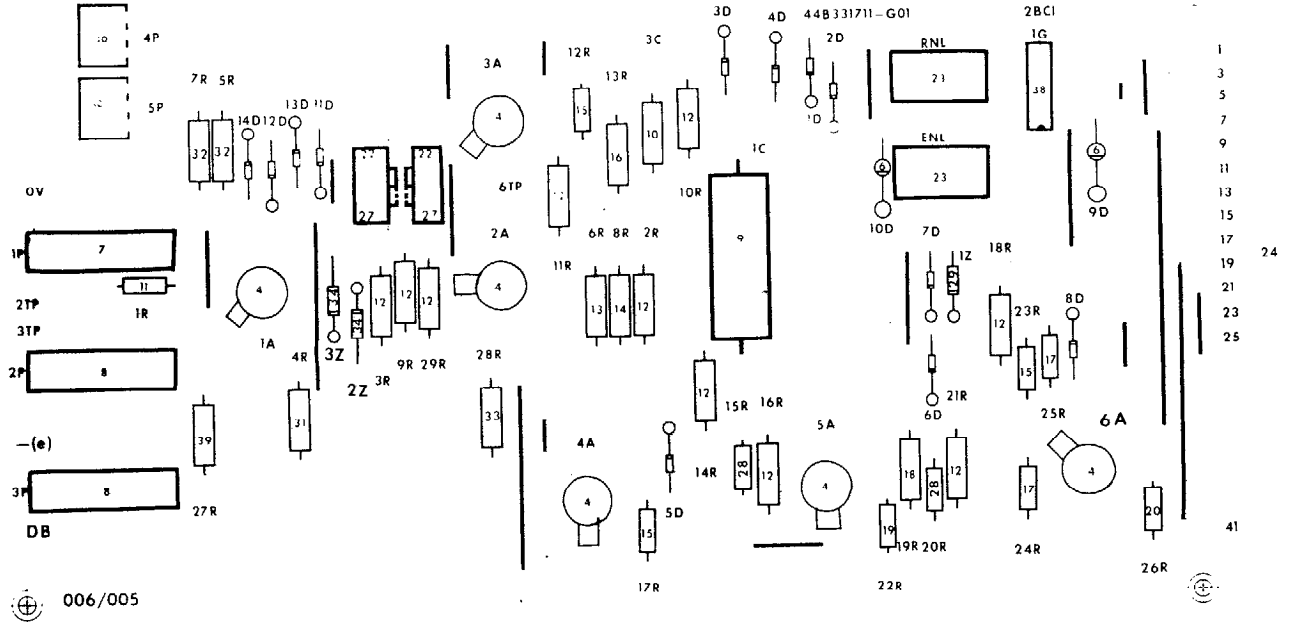
ASSEMBLY DRAWING

PL ISSUED

ASSEMBLY NOTES:

1. MASK OFF TEST POINTS, POTS, TERMINAL BOARD AND CONTACT TERMINALS PRIOR TO CONFORMAL COATING.
2. COAT BOARD PER PROCESS P60-HW8.
3. JUMPERS: USE #20 INSULATED WIRE.
4. IMPORTANT: INSTALL ITEM 25 AND 26 ONLY AFTER CLEANING AND CONFORMAL COATING.

PROTOTYPE 44B302099/006/005



COMPONENT SIDE VIEW (006)

REV		PRINTS TO
5	ANTI-9108 Sept. 21, 1972 W. Rowzie	11
6	ANTI-9608 Feb. 22, 1973 W. Lunsford	

RE: G. HOWDYSHELL 3/18/72
D K. Moyer Sept. 21, 1972

COMM & CONT PROD DEV
WAYNESBORO, VA.

44B331711 FL 1

2

GENERAL ELECTRIC

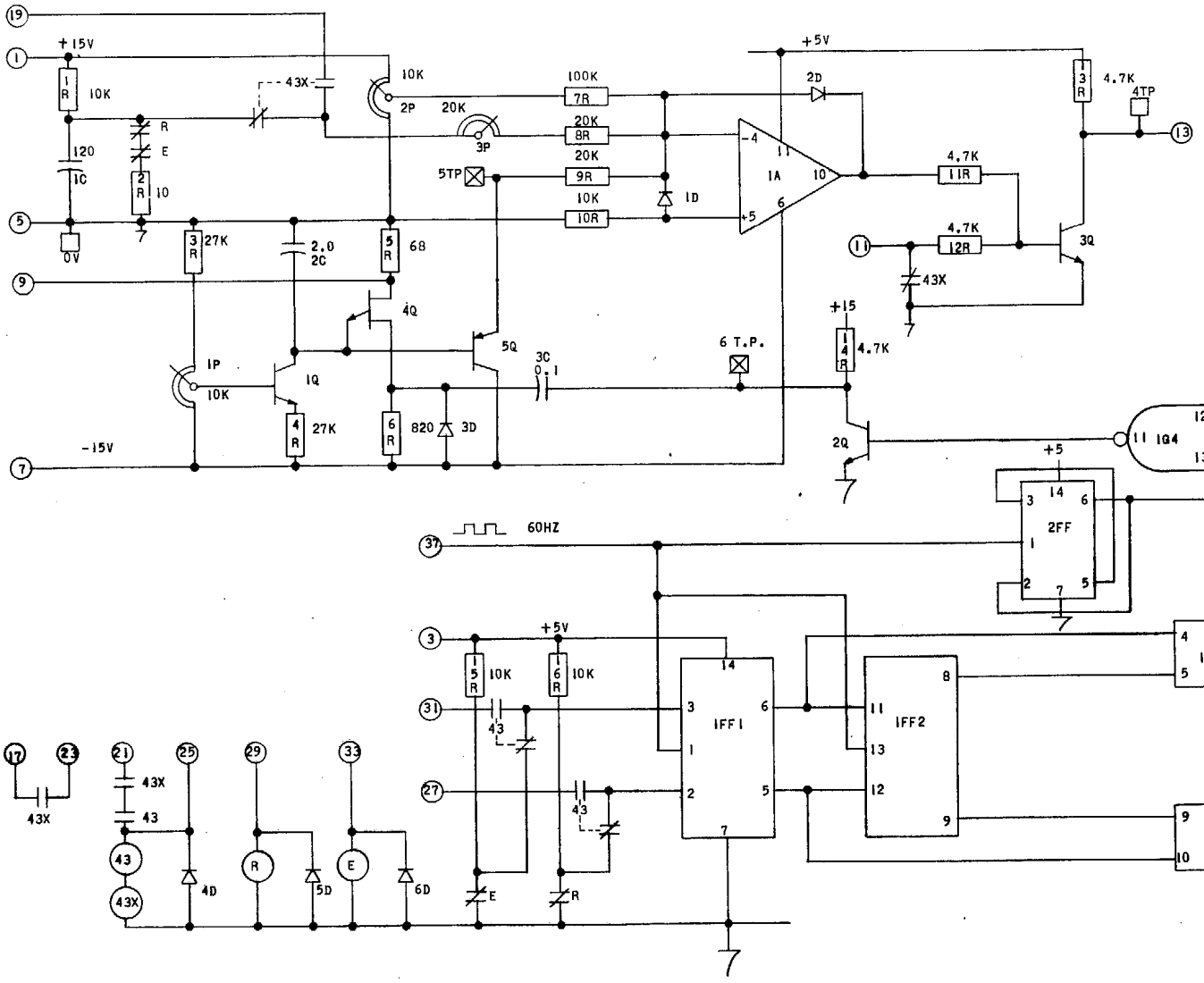
44C304809 CONT ON SHEET SH NO.

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE 2TRM

FIRST MADE FOR



1A	44A332168-002
2FF, 1FF	44A333402-006
1G	44A333402-001
1,2,3D	44B232028-001
4,5,6D	44B232019-004
1,2P	44A333423G10
3P	44A333423G20
43,43X	3SBH5008K1
R,E	3SAV5004M1
1,15,16R	933B590G10
2R	933B590D10
3,4R	933B590G27
5R	933B590D68
6R	933B590E82
7R	128A4057H10
8,9R	128A4057G20
10R	128A4057G10
11,12,13,14R	933B590F47
TEST POINTS	44B412208-001
1C	44B310008L10
2C	933B597J20
3C	933B597H10
1,2,3Q	44B311057-001
4Q	44A318523-001
5Q	44B238203-001

EXT. ADJ:

1P - TIME BASE FREQ.
2P - BIAS
3P - INPUT GAIN

TEST POINTS:

1TP - DIRECTION
2TP - DIRECTION
4TP - RUN SIGNAL (✓ = RUN; 0 = STOP)
5TP - TIME BASE
6TP - SYNC SIGNAL
0V - ZERO VOLTS

REVISIONS		PRINTS TO
1	E.A. Ellison June 27, 1973	11

MADE BY	APPROVALS			
E Ellison 1/28/71	HBC	COMM. & CONT. DEVICES WAYNESBORO, VA.	DIV OR DEPT LOCATION	44C304809 CONT ON SHEET SH NO.
ISSUED E Lotts Nov. 19, 1971				

GENERAL ELECTRIC

44B331720 FL 1

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:— APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINE DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

REV NO.

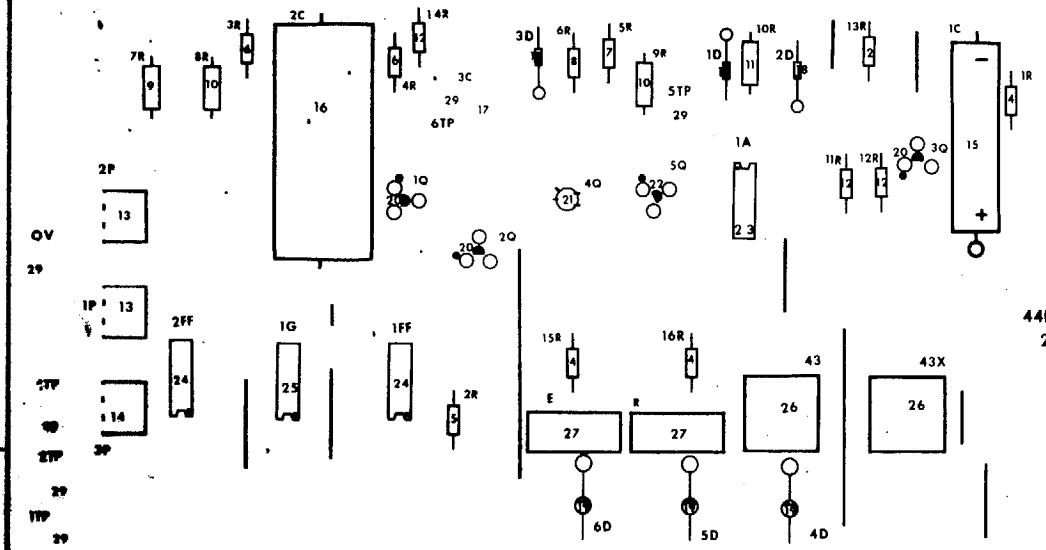
TITLE: PRINTED WIRING BOARD

2TRM & MANUAL

USED FOR 44A301995 (F.C.F.

ASSEMBLY DRAWING

PL ISSUED



44B304606/006/005

44B331720-G01
2TRM&MANUAL

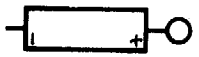
COMPONENT SIDE VIEW

ASSEMBLY NOTES:

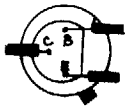
1. PARTS LIST ITEM NUMBER IS SHOWN INSIDE COMPONENT SYMBOL.
2. INSTALL DIODES WITH CATHODE END TOWARD CIRCLED PAD.



3. INSTALL CAPACITORS WITH POSITIVE END TOWARD CIRCLED PAD.

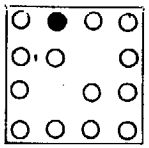


4. INSTALL CHIPS ORIENTED AS SHOWN.
5. INSTALL TRANSISTORS WITH FLAT TOWARD TAB.

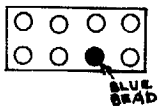


TOP VIEW

6. MOUNT RELAYS AS SHOWN BELOW. POSITION BLUE BEAD AS INDICATED.



TOP VIEW







7. TRANSISTOR 4Q & 5Q MUST BE PLACED AS SHOWN HERE.
8. MASK OFF TEST POINTS 1TP, 2TP, 0V, 4TP) AND POTS 1P, 2P, 3P PRIOR TO CONFORMAL COATING.
9. COAT BOARD PER PROCESS P6C-HW8.
10. IMPORTANT: INSTALL ITEM 30 AND 31 ONLY AFTER CLEANING AND CONFORMAL COATING.

REVISIONS			PRINTS TO
3	W. Lunsford	May 19, 1972	11
	ANTZ-9327		
4	ANTZ-9716 R. Kinzie	Sept. 21, 1972	
5	ANTZ-9276 W. Lunsford	Nov. 27, 1972	
6	ANTZ-9600 W. Lunsford	Feb. 27, 1973	

MADE BY: A. Pompeo 4-14-71
DESIGNED BY: D. Moyer May 17, 1972

RELEASED BY: R.M.B. COMM & CONT PROD DEV, WAYNESBORO, VA.

44B331720 FL 1

2

GENERAL ELECTRIC

44C304807 CONT ON SHEET SH NO.

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

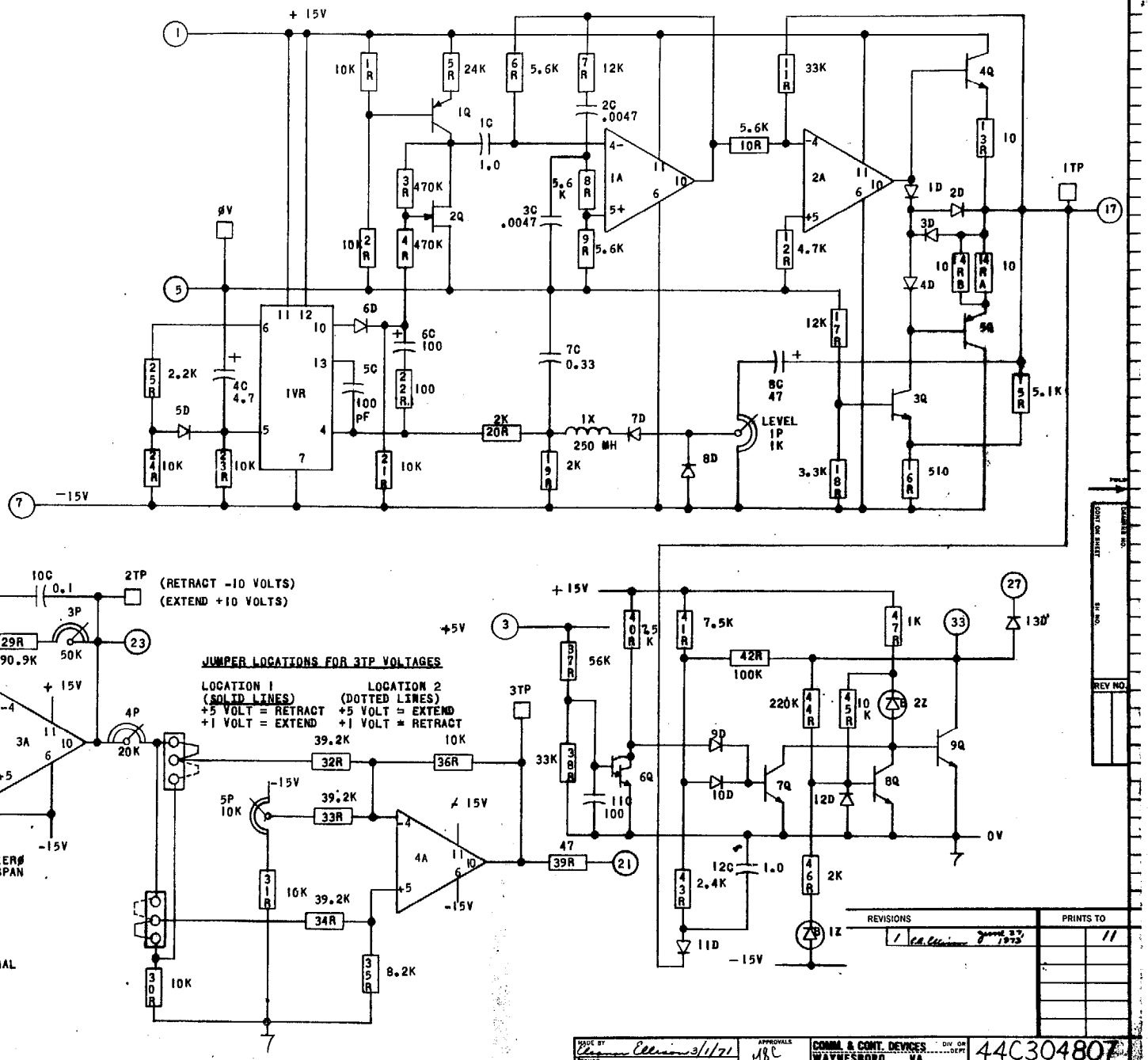
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS ±	DECIMALS ±	ANGLES ±
	✓			

TITLE

20SC

FIRST MADE FOR

1A-4A	44A332168-002
1VR	44A333413-001
1Q	44B238203-001
2Q	44A316720-001
3,7,8,9Q	44B311057-001
4Q	44A319819-001
5Q	44A417018-001
1-12D	44B232028-001
1C	933B597J10
2,3C	933B597F47
4C	44B310007J47
5C	933B599E10
6C,11C	44B310007L10
7C	933B597H33
8C	44A417046K47
1P	44A333423F10
1,2,21, 23,24R, 30R, 45R	933B590G10
3,4R	933B590H47
5R	933B590G24
6,8,9,10R	933B590F56
7,17R	933B590G12
11,38R	933B590G33
12R	933B590F47
13, 14RA, 14RB	933B590D10
15R	933B590F51
16R	933B590E51
18R	933B590F33
46, 20R	933B590F20
22R	933B590E10
19R	128A4057F20
25R	933B590F22
1X	44B311237-011
2P, 4P	44B318618G20
3P	44B318618G50
5P	44B318618G10
26R	128A4057H887
27,31,36R	128A4057G10
28R	933B590F91
29R	128A4057G909
32,33,34R	128A4057G392
35R	933B590F82
39R	933B590D47
37R	933B590G56
40,41R	933B590F75
42R	933B590H10
43R	933B590F24
44R	933B590H22
47R	933B590F10
9,10C	933B597H10
12C	44A417086J10
TERM. BD.	44B310014-201
6Q	44A315973-001
13D	44B232011-001
1,2Z	IN960B
U-CLIP	44A219567-001
TEST POINT	44B412208-001



REVISIONS

1	C.A. Ellenson	June 23, 1971	

PRINTS TO 11

MADE BY	DATE	APPROVALS	
C. Ellenson	3/11/71	MBC	
ISSUED C. Zotto	Nov. 19, 1971		

COMML & CONT. DEVICES DIV OR DEPT
WAYNESBORO, VA. LOCATION

44C304807 CONT ON SHEET SH NO. 1

DRAWING NO. 44B331722 SH NO.
CONT ON SHEET

GENERAL ELECTRIC

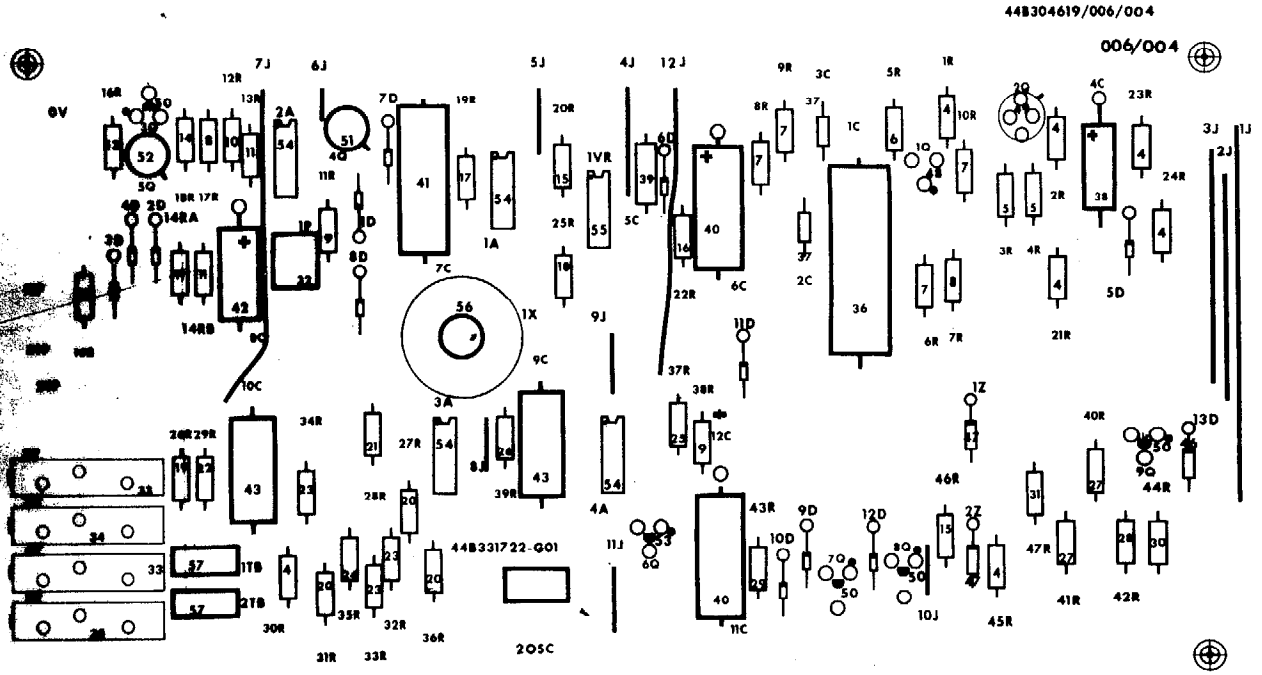
44B331722
CONT ON SHEET FL SH NO 1

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—					REV NO
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS			
		FRACTIONS	DECIMALS	ANGLES	
	✓	±	±	±	CONT ON SHEET NO

TITLE
PRINTED WIRING BOARD
20SC
FIRST MADE FOR (F.C.F.

ASSEMBLY DRAWING

PL ISSUED



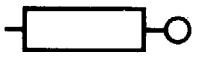
COMPONENT SIDE VIEW

ASSEMBLY NOTES:

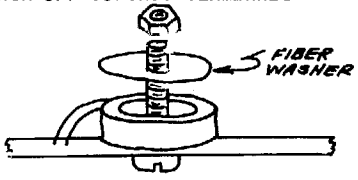
1. PARTS LIST ITEM NUMBER IS SHOWN INSIDE COMPONENT SYMBOL.
2. INSTALL DIODES WITH CATHODE END TOWARD CIRCLED PAD.



3. INSTALL CAPACITORS WITH POSITIVE END TOWARD CIRCLED PAD.



4. INSTALL TRANSISTORS AS SHOWN.
5. INSTALL 1X, ITEM 56, USING #4 SCREW AND NUT, USE FIBER WASHER BETWEEN 1X AND NUT.
6. A. COAT BOARD PER P6C-HW8 MASK OFF THE ADJUSTING SCREW ON 1P, 2P, 3P, 4P & 5P.
   B. MASK OFF 1TP, 2TP, & 3TP, 0V.
   C. MASK OFF CONTACT TERMINALS.



7. IMPORTANT: INSTALL ITEM G1 AND G2 ONLY AFTER CLEANING AND CONFORMAL COATING.

REVISIONS		PRINTS TO
F	AN72-3105 Sept 21, 1972 A. Ramos	11
G	AN73-7000 W. Linnard Feb 23, 1973	

MADE BY JERRY W STALLINGS 4-15-71
RE ISSUED D.K Moyer Sept. 21, 1972
RE APPROVED R.M.B
COMM & CONT PROD DEV
WAYNESBORO, VA.

44B331722
CONT ON SHEET FL SH NO 1

FE-806-F (9-70)
PRINTED IN U.S.A.

2

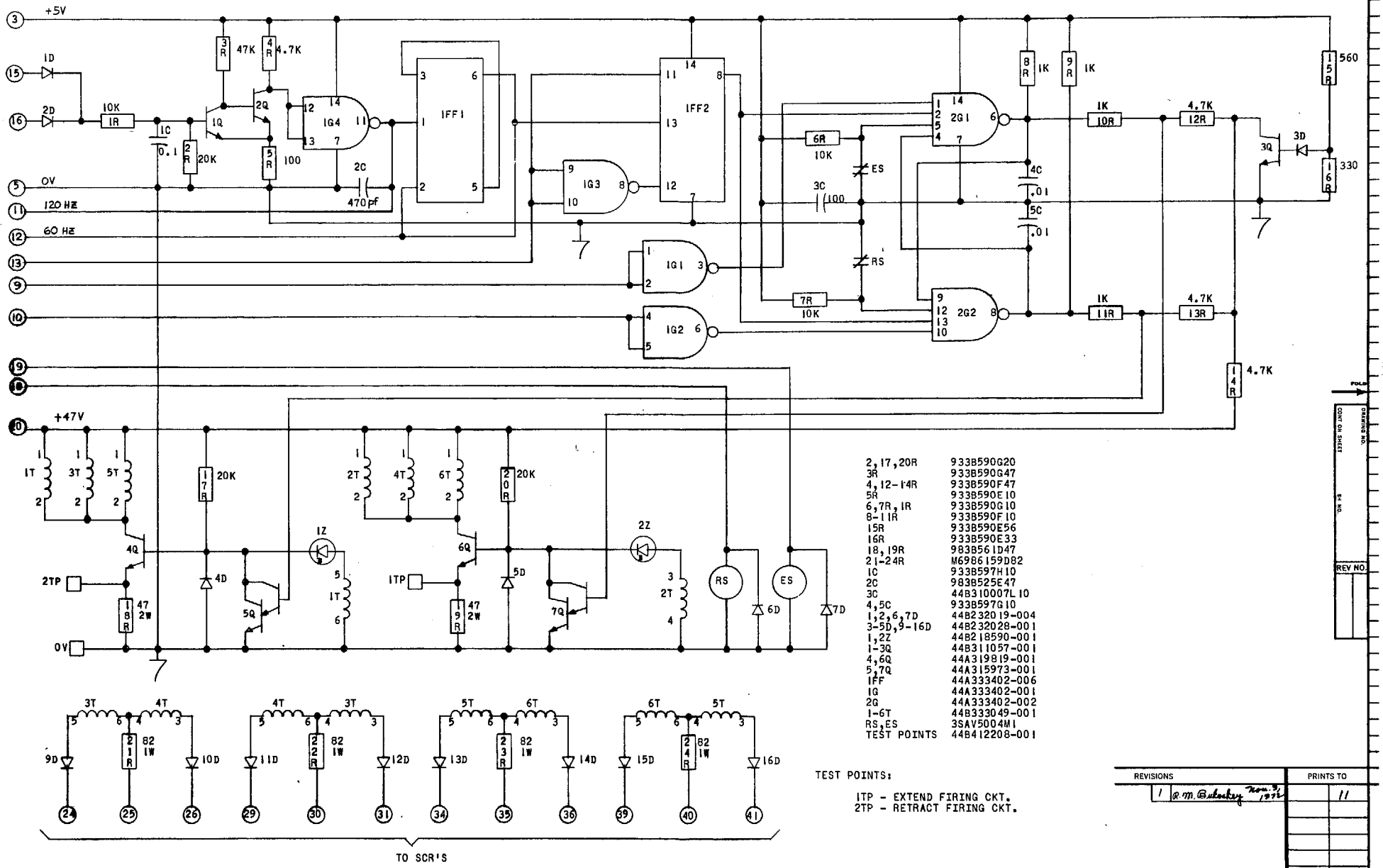
GENERAL ELECTRIC

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—				
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS		
		FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE: 2 SCR

FIRST MADE FOR

NOTES
1. RESISTOR VALUES ARE IN OHM AND ARE 1/2 WATT UNLESS OTHERWISE SPECIFIED.
2. CAPACITOR VALUES ARE IN MFD. UNLESS OTHERWISE SPECIFIED.



2,17,20R	933B590G20
3R	933B590G47
4,12-14R	933B590F47
5R	933B590E10
6,7R,1R	933B590G10
8-11R	933B590F10
15R	933B590E56
16R	933B590E33
18,19R	983B561D47
21-24R	M6986159D82
1C	933B597H10
2C	983B525E47
3C	44B310007L10
4,5C	933B597G10
1,2,6,7D	44B232019-004
3-5D,9-16D	44B232028-001
1,2Z	44B218590-001
1-3Q	44B311057-001
4,6Q	44A319819-001
5,7Q	44A315973-001
1FF	44A333402-006
1G	44A333402-001
2G	44A333402-002
1-6T	44B333049-001
RS,ES	3SAV5004M1
TEST POINTS	44B412208-001

TEST POINTS:

1TP - EXTEND FIRING CKT.
2TP - RETRACT FIRING CKT.

REVISIONS	PRINTS TO
1 R.M. Bukosky Nov. 9, 1971	11

MADE BY	APPROVALS		
Clarence Ellerson 1/28/71	MB	COMM. & CONT. DEVICES DIV. OR DEPT. WAYNESBORO, VA. LOCATION	44C304808 CONT ON SHEET SH. NO.
ISSUED R. Rose Nov. 19, 1971			

DRAWING NO 44B331719 CONT ON SHEET SH NO

GENERAL ELECTRIC

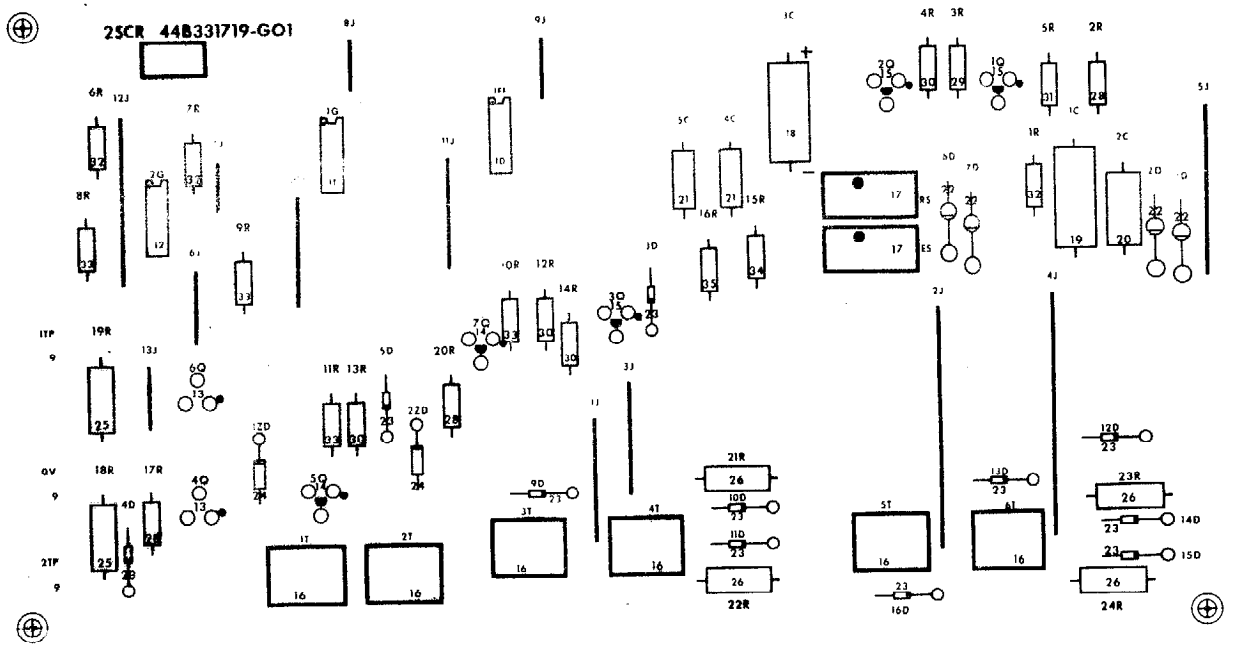
44B331719 CONT ON SHEET FL SH NO 1

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—					REV NO.	TITLE
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS				ETCHED WIRING BOARD
		FRACTIONS	DECIMALS	ANGLES		
	✓	±	±	±	CONT ON SHEET SH NO	FIRST MADE FOR 44A 2SCR

ASSEMBLY DIAGRAM

COMPONENT SIDE (/00

44B304601/006/001



1. COAT BOARD PER P6C-HW8.
2. MASK ITEMS 5 & 6, UPPER AND LOWER CONTACTS, AND ITEM 9, TEST POINTS, BEFORE COATING BOARD.

11,13

JUMPER TABLE

JUMPER No.	LGTH.	QTY.
1J	1.1	1
3J, 10J, 11J	1.0	3
9J, 13J	0.6	2
7J, 8J	0.5	2
6J	0.7	1
5J, 12J	1.5	2
2J, 4J	2.1	2

3. IMPORTANT: INSTALL ITEM 7 AND 8 ONLY AFTER CLEANING AND CONFORMAL COATING.

REVISIONS		PRINTS TO
1	W.h.h. Dec. 2, 1971 AN71-4081	11
2	W.h.h. Apr. 20, 1972 AN72-0238 Pumpkin	
3	73-9600 Feb. 23, 1973	

MADE BY J.W. STALLINGS 3/12/71 ISSUED

RE APPROVALS R.M.B

COMM. & CONTROL DIV OR DEPT

WAYNESBORO, VA. LOCATION

44B331719 CONT ON SHEET FL SH NO 1

2

FOLD

FOLD

PF-000-F (12-67) PRINTED IN U.S.A.

GENERAL ELECTRIC

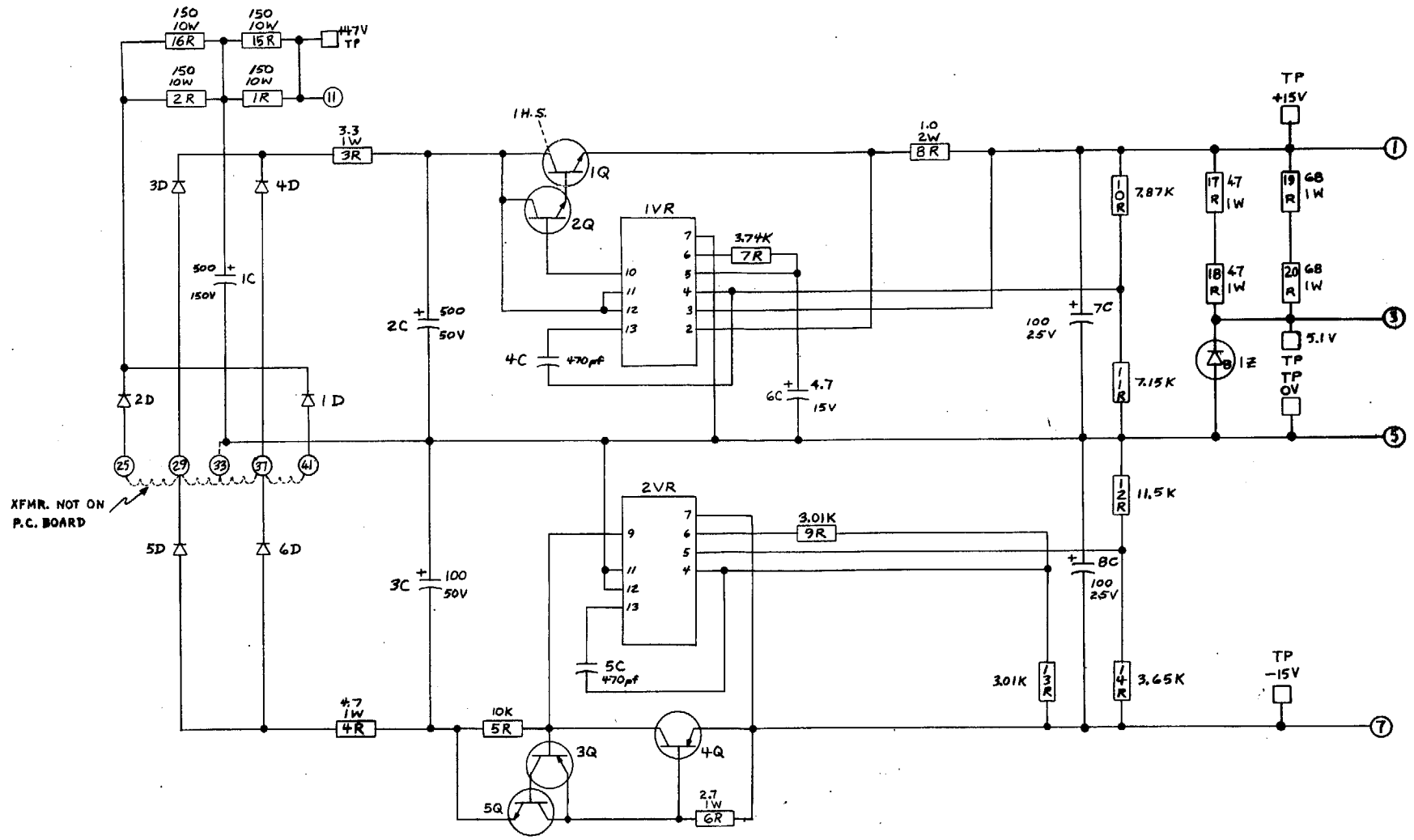
44C304149

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—				
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS		
		FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE: POWER SUPPLY

FIRST MADE FOR: ELEM. DIAG.

1,2,15,16R	K8622212E15	12R	128A4057G115		
3R	983B561C33	14R	128A4057F365		
4R	M6986159C47	17,18R	M6986159D47		
5R	933B590G10	19,20R	M6986159D68		
6R	M6986159C27	1C	K9774720L50	1D-6D	44B232019-004
7R	128A4057F374	2C	K9774719L50	1Z	IN3826A
8R	983B561C10	3C	K9774719L10	1,5Q	2N3054
9,13R	128A4057F301	4,5C	933B599E47	2Q	44B311057-001
10R	128A4057F787	6C	44B310007-J47	3,4Q	44B238246-001
11R	128A4057F715	7,8C	44B310008-L18	1,2VR	44A333413-001



NOTES: 1) RESISTOR VALUES ARE IN OHMS AND ARE 1/2W UNLESS OTHERWISE SPECIFIED.

2) CAPACITOR VALUES ARE MFD. UNLESS OTHERWISE SPECIFIED.

REVISIONS		PRINTS TO
1	W.R. Bergeron NOV 19	11
2	R.J. Quaid NOV. 9, 1971	
3	E.A. Allison June 27, 1972	

MADE BY E. Allison 10/19/70 ISSUED OCT 19 1970

APPROVALS MBL

COMM. & CONT. DEVICES DIV. OR DEPT. WAYNESBORO, VA. LOCATION

44C304149 CONT ON SHEET

FP-0004-P 10-69 PRINTED IN U.S.A.

GENERAL ELECTRIC

44B331715 CONT ON SHEET All SH NO. 1

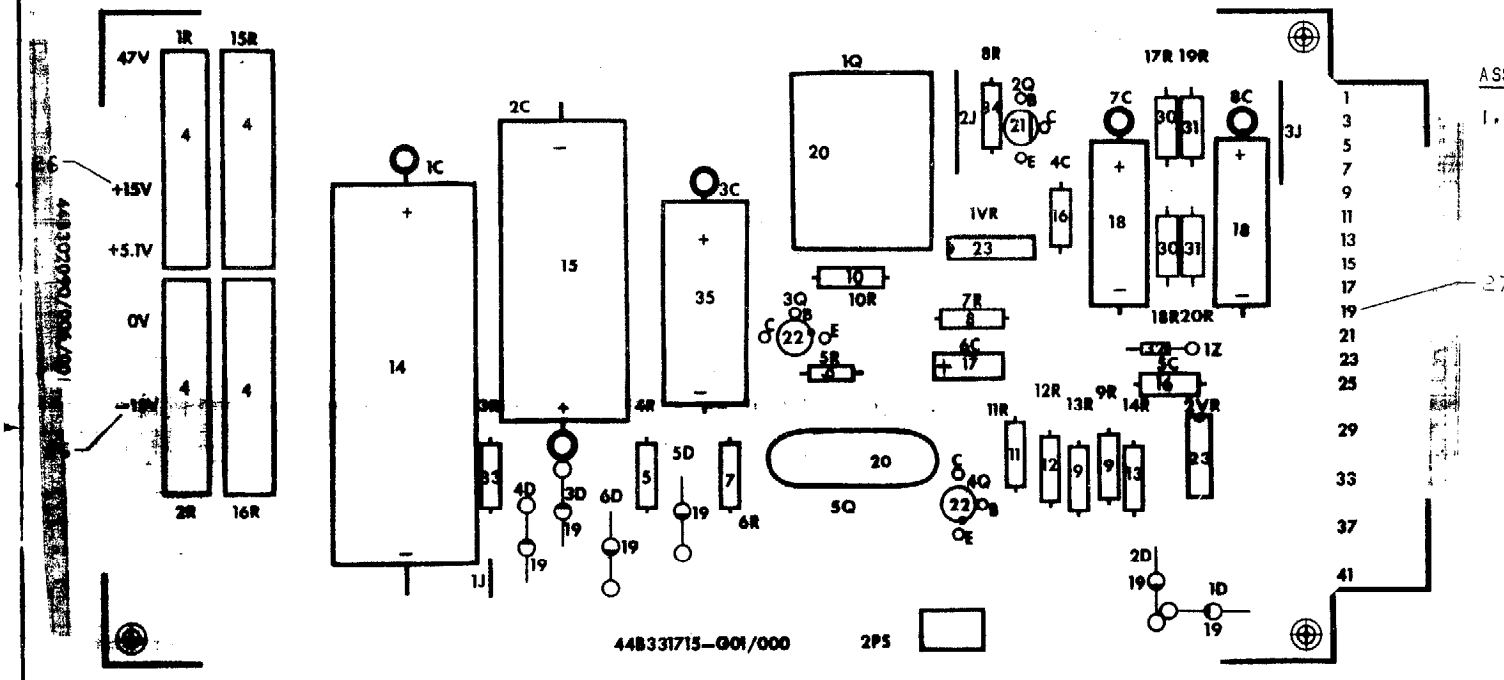
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—				
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS		
		FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

REV NO. | CONT ON SHEET | SH NO

TITLE

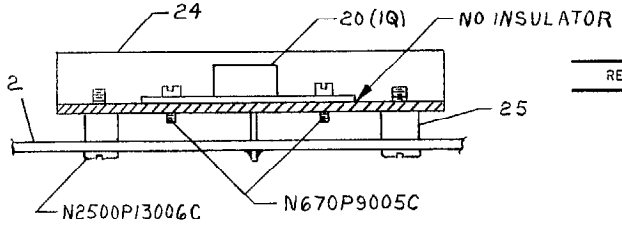
FIRST MADE FOR 44A301979 2Ps

Assembly Dwg



ASSEMBLY NOTE:

1. IMPORTANT: INSTALL ITEM 28 AND 29 ONLY AFTER CLEANING AND CONFORMAL COATING.



1. PARTS LIST ITEM NUMBER IS SHOWN INSIDE COMPONENT SYMBOL.
2. INSTALL DIODES WITH CATHODE END TOWARD CIRCLED PAD.
3. INSTALL CAPACITORS WITH POSITIVE END TOWARD CIRCLED PAD.
4. MASK TEST POINTS & CONTACT TERMINALS PRIOR TO CONFORMAL COATING.
5. COAT BOARD PER PROCESS P6C-HW8.
6. TIE DOWN CAPACITORS 1C, 2C, 3C TO PRINTED CIRCUIT BOARD.

REVISIONS			PRINTS TO
1	AN 72-8171 A. Chapelle	Mar. 16 1972	11
2	AN 73-7554 W. Lunsford	Jan. 26, 1973	
3	AN 73-7800 W. Lunsford	Feb. 23 1973	

DRAWN BY	APPROVALS			
C. Greaves 2-3-71	R. M. F.	COMM & CONT DEVICES DEPT	44B331715	
P. Clifton Sept. 28, 1971		WAYNESBORO, VA. LOCATION	CONT ON SHEET All	SH NO. 1

FF-806-F (6-69) PRINTED IN U.S.A.

2

GENERAL ELECTRIC

44C304455 CONT ON SHEET SH NO

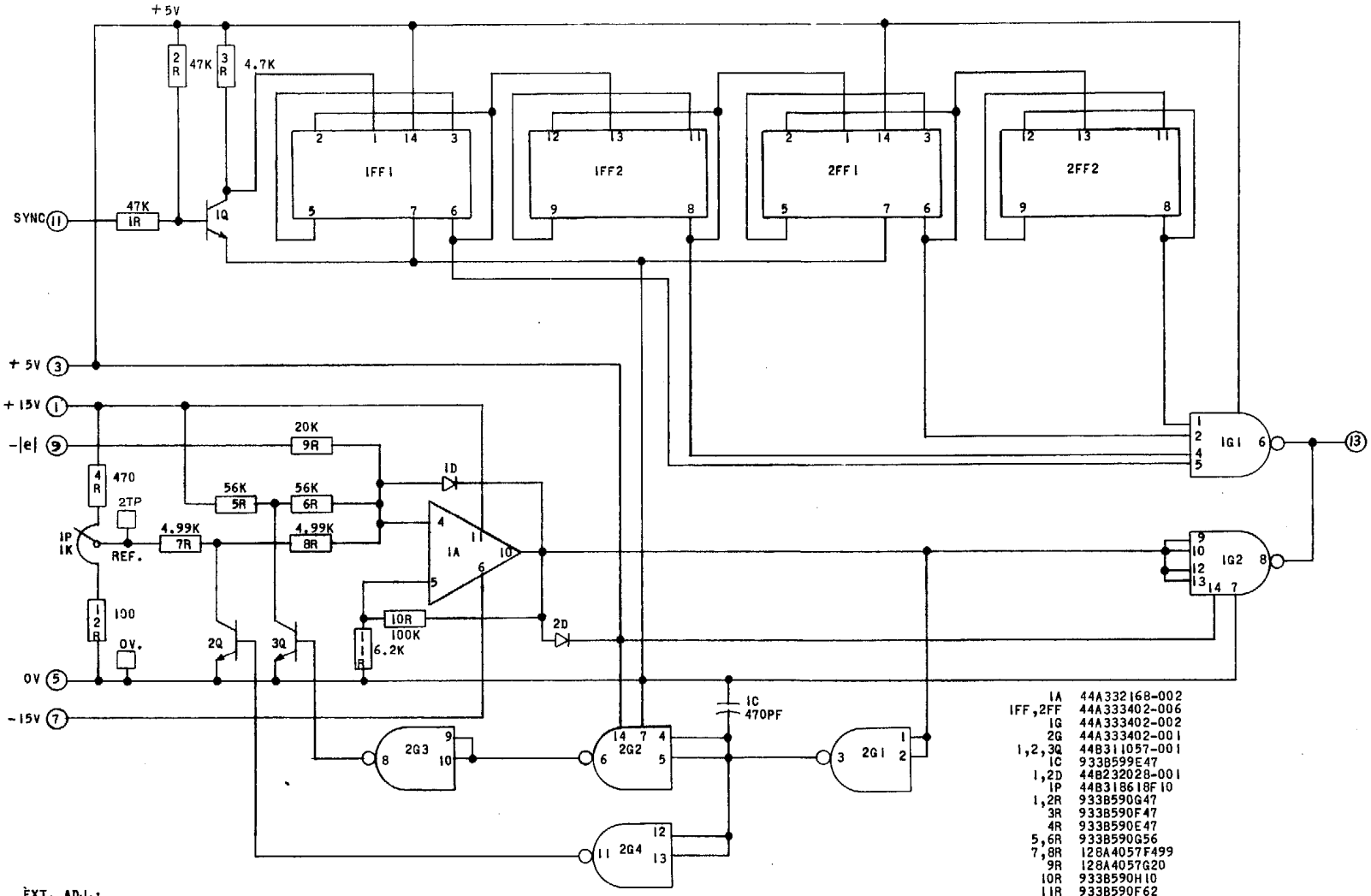
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINE DIMENSIONS: FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE

ADF

FIRST MADE FOR



1A	44A332168-002
1FF,2FF	44A333402-006
1G	44A333402-002
2Q	44A333402-001
1,2,3Q	44B311057-001
1C	933B599E47
1,2D	44B232028-001
1P	44B318618F10
1,2R	933B590G47
3R	933B590F47
4R	933B590E47
5,6R	933B590G56
7,8R	128A4057F499
9R	128A4057G20
10R	933B590H10
11R	933B590F62
	44B412208-001 TEST POINTS
12R	933B590E10

EXT. ADJ.:

1P - OPERATE LEVEL

TEST POINTS:

0V - ZERO VOLTS
2P - OPERATE LEVEL VOLTS

DRAWING NO 44C304455 CONT ON SHEET SH NO

REV NO.

REVISIONS		PRINTS TO
1	R.M. Bukosky Nov. 9, 1971	11
2	R.M. Bukosky June 6, 1972	

MADE BY Norman Ellison 1/20/71
ISSUED C. Botto Nov. 23, 1971
APPROVALS MSC
COMML & CONT. DEVICES DIV OR DEPT
WAYNESBORO, VA. LOCATION
44C304455 CONT ON SHEET SH NO

GENERAL ELECTRIC

44B331712 CONT ON SHEET FL SH NO 1

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

REV

CONT ON SHEET SH NO

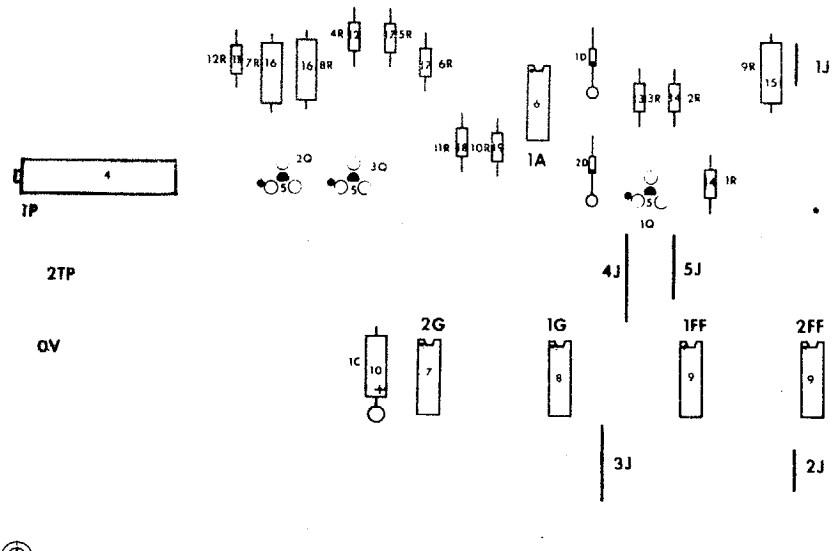
TITLE **ETCHED WIRING BOARD**

ADF

FIRST MADE FOR 44A301992

ARTWORK

COMPONENT SIDE (/006)



44B304650/006/001

44B331712-G01/001 ADF

006/001

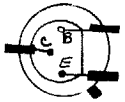
1. PARTS LIST ITEM NUMBER IS SHOWN INSIDE COMPONENT SYMBOL.
2. INSTALL DIODES WITH CATHODE END TOWARD CIRCLE PAD.



3. INSTALL CAPACITORS WITH POSITIVE END TOWARD CIRCLE PAD.



4. INSTALL CHIPS ORIENTED AS SHOWN
5. INSTALL TRANSISTORS WITH FLAT EDGE TOWARD TAB.



6. NO PROJECTION GREATER THAN .70" FROM THE SURFACE OF THE COMPONENT SIDE OF THE BOARD IS PERMITTED.
7. NO PROJECTION GREATER THAN .70" FROM THE SURFACE OF THE SOLDER SIDE OF THE BOARD IS PERMITTED.
8. A. COAT BOARD WITH CONFORMAL COATING PER P6C-HW8.
   B. MASK OFF ADJUSTING SCREW ON 1P.
   C. MASK OFF 2TP, 0V, AND CONTACT TERMINALS
9. IMPORTANT: INSTALL ITEM 28 AND 29 ONLY AFTER CLEANING AND CONFORMAL COATING.

Jumper Table

Jumper No	Length	Qty
1J, 2J	.60	2
3J	.80	1
4J	.90	1
5J	.70	1

REVISIONS		PRINTS TO
1	W.L.H. Sept. 13, 1971 7/-403B	11
2	W. Lunsford April 20, 1972 72-823B	
3	W.T.S. 7-20 W. Lunsford Feb. 22, 1973	

MADE BY R. Pompeo 2/1/71

RELEASED P. Chittum Sept. 13, 1971

APPROVAL R M B

COMM. & CONTROL DIV OR DEPT

WAYNESBORO, VA. LOCATION

44B331712 CONT ON SHEET FL SH NO 1

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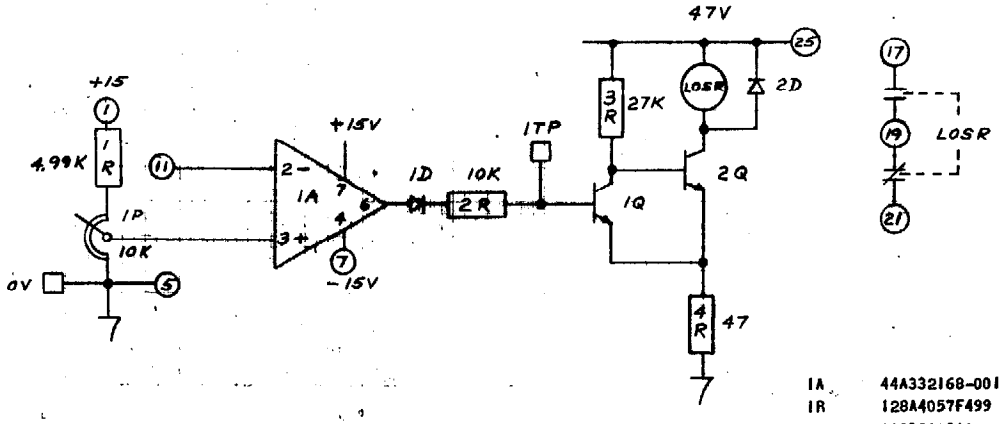
GENERAL ELECTRIC

44C305892

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—				
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS		
		FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE: LOS

FIRST MADE FOR



1A	44A332168-001
1R	128A4057F499
2R	933B590G10
3R	933B590G27
4R	933B590D47
IP	44B318618-G10
ID	44B232028-001
2D	44B232019-004
1,2Q	44B311057-001
LOSR	3SAN5004M1

44C305892 REV NO.

REVISIONS			PRINTS TO
1	R M Cubley	Nov. 3, 1972	11
2	C.A. Ellis	Dec. 15, 1972	
3	C.A. Ellis	June 27, 1973	

MADE BY: C. Ellis 4/19/72 — ISSUED: D. Yerger May 16, 1972

APPROVALS: R.M.C.

DRIVE SYSTEMS PROD. DEPT. — WATHESBORO, NC. — LOCATION

44C305892 — DWG NO. — SH NO.

1

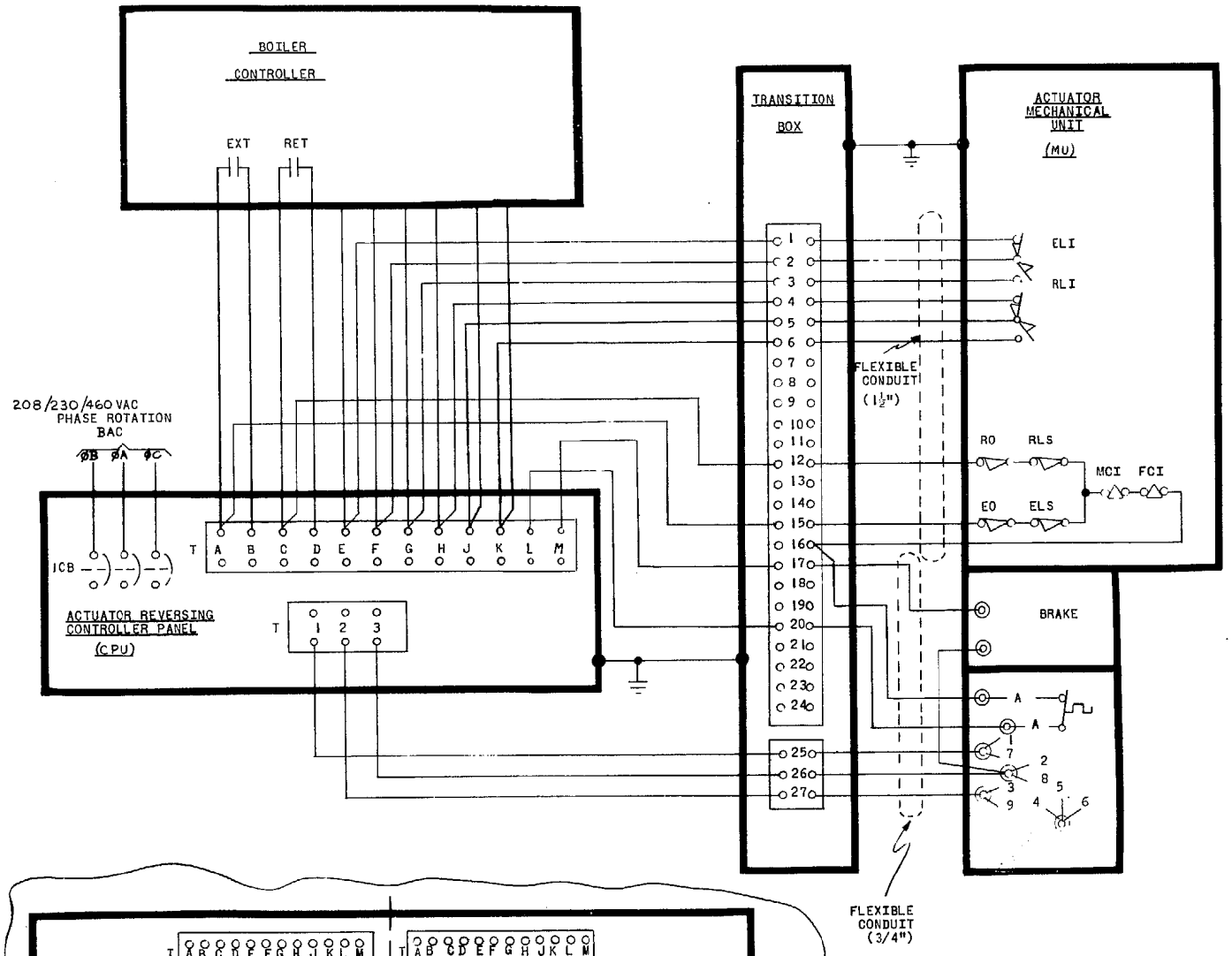
GENERAL ELECTRIC

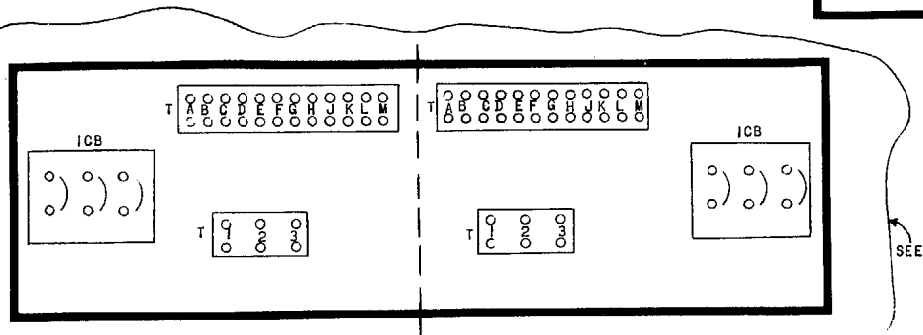
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—				
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS ±	DECIMALS ±	ANGLES ±
	✓			

TITLE

A.C. ACTUATOR INTERCONNECTION
NON-MODULATING CONTROL

FIRST MADE FOR





BACK VIEW - CPU

FIGURE 1

NOTES:

1. FLEXIBLE CONDUITS & TRANSITION BOX ARE STANDARD EQUIPMENT. EACH CONDUIT SUPPLIED IN AN 8 FT LENGTH.
2. FOLLOWING APPLIES TO A.C. ACTUATOR SIZES 2,3, & 4.
   - A. SUPPLIED INTERCONNECTING CABLE:
     FROM TRANSITION BOX TO MECHANICAL UNIT
     1) POWER - 12 AWG.; QTY. 3
     2) CONTROL - 16 AWG.; QTY. 12
     3) GND. WIRE - 14 AWG.; QTY. 1
   - B. REQUIRED INTERCONNECTING CABLE:
     FROM TRANSITION BOX TO REVERSING CONTROLLER
     1) POWER - 12 AWG.; QTY. 3
     2) CONTROL - 16 AWG.; QTY. 10
     3) GND. WIRE - 14 AWG.; QTY. 1
   - C. REQUIRED INTERCONNECTING CABLE:
     FROM BOILER CONTROLLER TO REVERSING CONTROLLER
     1) CONTROL - 16 AWG.; QTY. 8
   - D. REQUIRED INTERCONNECTING CABLE:
     FROM CUST. 3Ø SOURCE TO CIRCUIT BREAKER
     1) POWER - 12 AWG.; QTY. 3
3. FOLLOWING APPLIES TO A.C. ACTUATOR SIZE 5
   - A. SUPPLIED INTERCONNECTING CABLE:
     FROM TRANSITION BOX TO MECHANICAL UNIT
     1) POWER - 10 AWG.; QTY. 3
     2) CONTROL - 16 AWG.; QTY. 12
     3) GND. WIRE - 14 AWG.; QTY. 1
   - B. REQUIRE INTERCONNECTING CABLE
     FROM TRANSITION BOX TO REVERSING CONTROLLER
     1) POWER - 20 AWG.; QTY. 3
     2) CONTROL - 16 AWG.; QTY. 10
     3) GND. WIRE - 14 AWG.; QTY. 1
   - C. REQUIRED INTERCONNECTING CABLE:
     FROM BOILER CONTROLLER TO REVERSING CONTROLLER
     1) CONTROL - 16 AWG.; QTY. 8
   - D. REQUIRED INTERCONNECTING CABLE:
     FROM CUSTOMER 3Ø SOURCE TO CIRCUIT BREAKER
     1) POWER - 10 AWG.; QTY. 3
4. TERMINALS TA THROUGH TK ARE ONLY TIE POINTS FOR LIMIT SWITCH INDICATING CONTACTS ELI & RLI.
5. SOME CONTROLLER PANELS MAY CONTAIN 2 COMPLETE BUT SEPARATE CIRCUITS. THE BACK VIEW OF THE PANEL WILL LOOK LIKE FIGURE 1. TERMINAL BOARDS AND A CIRCUIT BREAKER JUST TO THE LEFT OF THE DOTTED LINE ARE CONNECTION POINTS FOR A CIRCUIT CONTROLLING ONE MECHANICAL UNIT (MU). TERMINAL BOARDS AND A CIRCUIT BREAKER TO THE RIGHT OF THE DOTTED LINE ARE CONNECTION POINTS FOR A CIRCUIT CONTROLLING ANOTHER MECHANICAL UNIT(MU). CONNECTION POINTS TO EACH CIRCUIT ARE IDENTICAL THEREFORE, ONLY ONE COMPLETE INTERCONNECTION OF COMPONENTS IS SHOWN.

DRAWING NO 44C305754 CONT ON SHEET SH NO

REV NO.

REVISIONS			PRINTS TO
1	E.A. Ellison	May 15, 1972	11
2	K.M. BLOSKEY	Nov. 22, 1972	
3	C.A. Carof	Dec. 19, 1972	

MADE BY	APPROVALS	COMM. & CONT. DEVICES DIV OR DEPT	44C305754
ISSUED C. Scott Dec. 9, 1971	RMC	WAYNESBORO, VA. LOCATION	CONT ON SHEET SH NO

GENERAL ELECTRIC

11-73(150)

Drive Systems Product Department
General Electric Company, Waynesboro, Virginia 22980